

The First Six Months Decide: A Data-Driven Retention Strategy for a Texas auto insurer

Final Stage 1 — Data Analytics Competition

Team DataWizards

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2026

Executive Summary

A Texas auto insurer loses 11.7% of its customers to non-renewal or cancellation, which puts about USD 17.8M of annual premium (11.8% of the premium base) at risk. Using records of 160,000 policyholders, we combined statistical driver analysis, survival analysis, interpretable regression, a cross-validated machine-learning ensemble and doubly robust causal inference to answer three questions: who leaves, when they leave, and which actions are worth funding.

Churn is most strongly associated with tenure band, age band and length of residence; customers in the riskiest group of tenure band ('0-6m') churn at 48.8%, 4.2 times the average. Risk is concentrated in time: the churn hazard peaks in tenure interval [90, 180) days (1.16% per 30 days, about 11.8 times the average). The final model reaches an out-of-fold ROC-AUC 0.702 (95% CI 0.698–0.707); the top 20% of scored customers contain 51% of all churners (2.6× random targeting). A retention campaign aimed at the top 10% of customers by expected loss is estimated to return USD 590.8K net (ROI 1.5×), whereas random targeting of the same size yields USD -35K.

We recommend (1) First-year onboarding programme; (2) Life-stage tailored engagement; (3) Household-based cross-sell and life-event triggers. Each action should start as a randomised pilot; detecting a reduction of 3.5 pp requires about 2,481 customers per arm.

1. Introduction

A personal auto insurer in Texas has seen roughly one in nine policyholders leave and wants to understand who leaves, when, and which retention actions are worth funding. The dataset (public Kaggle auto-insurance churn data) contains policy, demographic and household attributes for 200,000 customers.

For an individual insurer, churn (non-renewal or cancellation) affects several parts of the business at once. It erodes the renewal premium base, leaves acquisition costs unrecovered when policies exit early, distorts the persistency assumptions used in pricing (Eling & Kochanski, 2013) and is often a signal of unmet needs or poor service (Brockett et al., 2008). The economics of retention are well documented: Reichheld and Sasser (1990) reported that reducing defections by 5% raised profits by 25% to 85% across several service businesses. Not every exit is equally avoidable, however: Gottlieb and Smetters (2021) found that forgetting to pay explained 37.8% of recent life-insurance lapses, so part of churn can be addressed through process design rather than price.

Prior research links lapse and churn to income shocks, contract age, product type and payment frequency (Eling & Kiesenbauer, 2014; Hein et al., 2020) and to renewal pricing and the service experience (Brockett et al., 2008). Methodologically, churn models should be judged by the profit of the campaigns they enable (Verbraken et al., 2013), and risk should be distinguished from responsiveness to an intervention (Ascarza, 2018). Appendix B reviews the methods literature in more detail.

This study answers 4 research questions (RQ):

- RQ1. Which customer, policy and service factors are associated with churn, and how strongly after controlling for the others?
- RQ2. When during the customer lifecycle is churn risk highest, and which segments carry the highest risk?
- RQ3. How accurately can churn be predicted, and which customers should be prioritised?
- RQ4. Which interventions are most likely to reduce churn, for whom, and what is their expected economic value?

2. Data and Methodology

2.1 Data and preparation

The training data contain 160,000 customers (plus 40,000 customers to be scored) and 20 original variables. The target is 'Churn', defined as an account that was suspended (policy cancelled or not renewed) (churn = 1); the churn rate is 11.7%, so models are evaluated with ROC-AUC, precision–recall and profit rather than accuracy.

Preparation was automated and audited: 5 type corrections and harmonisations were applied (for example numbers or ranges stored as text, date parsing and inconsistent category labels), 0 duplicate records were removed and impossible values were set to missing. Most importantly, acct suspd date year, acct suspd date month, acct suspd date (days since) and acct suspd date are recorded only around or after the churn event; such post-event variables leak the outcome, so they and all derived features were removed before any analysis (single-feature and missingness tests flagged them automatically). We engineered 12 insurance-specific features, such as tenure (years), tenure band, new customer (<12 months), age band and premium-to-income ratio.

2.2 Analytical framework

The analysis follows five layers (Figure B.1). (1) Univariate tests (χ^2 and Mann–Whitney U) with effect sizes, Information Value and Benjamini–Hochberg correction screen candidate drivers (Benjamini & Hochberg, 1995). (2) Because churn is a time-to-event outcome with censored observations, Kaplan–Meier curves, log-rank tests, a Cox

model, an accelerated failure time model and restricted mean survival time describe when customers leave (Kaplan & Meier, 1958; Cox, 1972). (3) A logistic regression gives adjusted odds ratios that managers can read directly. (4) 7 algorithms — CatBoost, Explainable boosting machine (EBM), Hist. gradient boosting, LightGBM, Logistic regression, Neural network (MLP) and XGBoost — were trained with stratified 5-fold cross-validation, tuned with Bayesian optimisation and combined by ensemble selection; tree ensembles remain the strongest family on tabular data (Grinsztajn et al., 2022), and SHAP values explain each prediction (Lundberg & Lee, 2017). (5) Doubly robust estimators (AIPW and the DR-learner) estimate the average and heterogeneous effects of controllable levers (Chernozhukov et al., 2018; Kennedy, 2023). Results are translated into money with customer lifetime value, the expected maximum profit measure (Verbraken et al., 2013) and a campaign simulation.

2.3 Validation

All reported performance figures are out-of-fold estimates; target-dependent encodings and hyperparameter search were fitted inside training folds only. Models are compared with DeLong tests for correlated AUCs (DeLong et al., 1988) with Holm correction, uncertainty is reported with bootstrap and DeLong confidence intervals, and calibration is checked with the Brier score and expected calibration error. Sanity checks support the validity of the pipeline: a model trained on shuffled labels scores an AUC of 0.503, and an adversarial classifier separates training from scoring data with AUC 0.50. Appendix B documents the full validation design.

3. Findings

3.1 Who churns? (RQ1)

The strongest correlates of churn are tenure band (IV 0.89): '0-6m' churns at 48.8%, 4.2× the average; age band (IV 0.04): '25-34' churns at 14.6%, 1.2× the average; length of residence (IV 0.01): '0-1' churns at 13.5%, 1.2× the average (Table 1). Several variables — length of residence, has children, college degree and curr ann amt — are statistically detectable but practically negligible (IV < 0.02), a reminder that with 160,000 records significance alone is not evidence of importance.

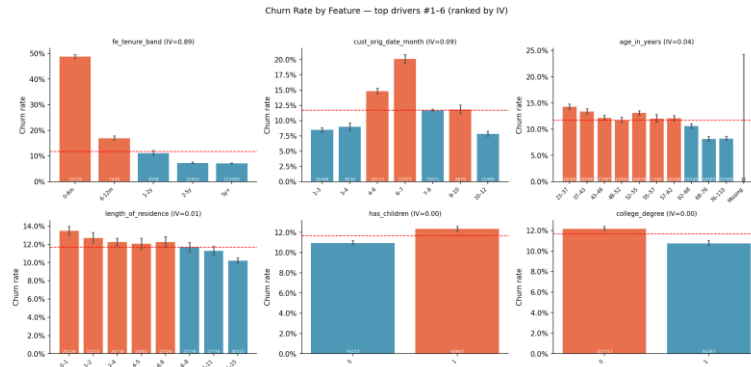


Figure 1. Churn rate by the strongest drivers (bars: churn rate with 95% Wilson CI; dashed line: overall churn rate).

Table 1. Evidence for churn drivers across methods (overall churn rate 11.7%; 'methods agreeing' counts significant univariate test, logistic regression, Cox model and top-10 SHAP).

Driver	Riskiest group	Churn rate	Lift	IV	Odds ratio	Hazard ratio	SHAP rank	Methods agreeing
tenure band	0-6m	48.8%	4.18×	0.89	12.20	—	10	3/4
age band	25-34	14.6%	1.25×	0.04	0.92	0.48	16	3/4
length of residence	0-1	13.5%	1.15×	0.01	—	—	4	2/4
college degree	No	12.2%	1.04×	0.00	—	—	5	2/4
has children	Yes	12.3%	1.06×	0.00	—	—	6	2/4
latitude	Missing	12.4%	1.07×	0.00	—	—	8	1/4
premium-to-income ratio	0.0182–0.0232	12.4%	1.06×	0.00	—	—	11	1/4

The multivariate logistic regression confirms that these effects are not artefacts of each other: holding the other variables constant, the largest adjusted effects are tenure band = 0-6m (vs '5y+': OR 12.20, 95% CI 11.72–12.71), tenure band = 6-12m (vs '5y+': OR 2.74, 95% CI 2.56–2.94) and tenure band = 1-2y (vs '5y+': OR 1.63, 95% CI 1.47–1.79) (full model in Appendix D).

3.2 When do customers churn? (RQ2)

Survival analysis treats active customers as censored observations. The Kaplan–Meier estimates show 98.6% after 90 days; 95.3% after 180 days; 94.4% after 365 days. The churn hazard peaks in tenure interval [90, 180) days at 1.16% per 30 days, against an average of 0.10% (Figure 2).

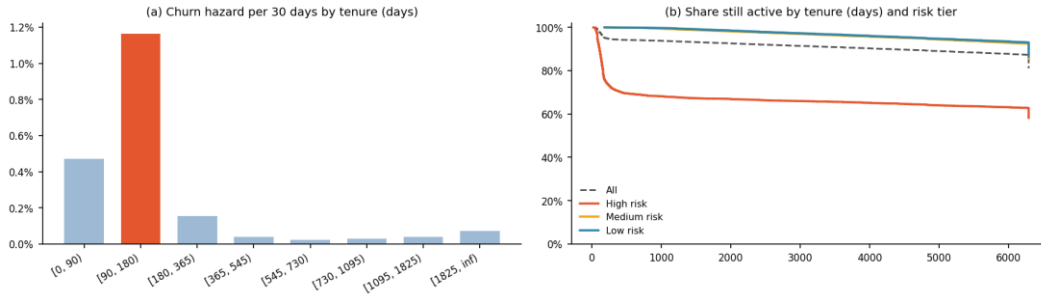


Figure 2. (a) Churn hazard per 30 days by tenure interval; (b) Kaplan–Meier survival by predicted risk tier (out-of-sample validation of the score over time).

Restricted mean survival time translates this into time lost: within the first 730 days, customers with age band = '<25' stay on average 57.9 days less than those in '65+'.

3.3 Who is most at risk? Interpretable segments

A shallow decision tree turns the drivers into segments operations teams can act on (Table F.1). The riskiest segment — tenure band = 0-6m AND cust orig date month ≥ 7 AND age in years ≤ 45 — has 5,105 customers (3.2% of the base) churning at 50.9% (4.4 $\times$) and contains 13.9% of all churners, whereas the most loyal segment churns at only 6.9%.

3.4 How well can churn be predicted? (RQ3)

Table 2. Out-of-fold performance (5-fold stratified CV; random baseline ROC-AUC 0.500, PR-AUC 0.117). EMPC = expected maximum profit per customer (Verbraken et al., 2013); p-values Holm-corrected.

Model	ROC-AUC (95% CI)	PR-AUC	Brier	EMPC (USD)	DeLong p vs final
Final ensemble: hill climb	0.702 (0.697–0.707)	0.329	0.088	4.31	—
LightGBM	0.700 (0.696–0.705)	0.325	0.088	4.31	0.210
XGBoost	0.700 (0.695–0.704)	0.328	0.090	4.23	0.149
xgb_multiseed	0.700 (0.695–0.704)	0.326	0.092	4.22	0.135
lgbm_tuned	0.699 (0.694–0.703)	0.326	0.090	4.25	0.141

The final model reaches an out-of-fold ROC-AUC 0.702 (95% CI 0.698–0.707) and a PR-AUC of 0.329, 2.8 times the random baseline (Table 2). The interpretable logistic regression reaches 0.696; the ensemble is significantly better (Δ AUC +0.006, DeLong p 0.013), so the added complexity buys real accuracy. A glass-box Explainable Boosting Machine reaches 0.693. Ranking by expected maximum profit differs from ranking by AUC (most profitable: Hist. gradient boosting). Concentration is strong: the highest-risk decile churns at 49.4% and the top 20% contain 51% of churners.

Model development was systematic: 30 logged experiments covered the model zoo, tuning, multi-seed averaging, feature selection, ablation and ensembling (Appendix F). An ablation study shows that removing tenure costs 0.023 AUC (DeLong $p < 0.001$), the largest contribution of any feature group.

3.5 Why is a customer at risk? Explaining the model

SHAP values attribute each prediction to the features (Lundberg & Lee, 2017). The model relies mostly on days tenure (52%), tenure (years) (15%) and cust orig date (days since) (7%) of total attribution (Figure G.4), consistent with the statistical evidence, so the model behaves in a way managers can audit.

4. Business Impact (RQ4)

4.1 Value at risk and prioritisation

Weighting each customer's churn probability by annual premium, about USD 17.8M of annual premium (11.8% of the premium base) is expected to be lost. Crossing risk with value (Figure 3b) isolates a 'Priority Save' group — 11.4% of customers with high risk and above-median premium — that holds 35% of the value at risk and warrants personal contact; high-risk, low-value customers can be served through automated digital touchpoints.

4.2 Retention campaign economics

We simulated a retention campaign on the out-of-fold predictions, assuming: a contact-and-incentive cost of USD 25 per customer, that 25% of contacted would-be churners are retained and a 15% margin over a 5-year horizon (discount rate 10%). Contacting the top 10% by expected loss maximises net benefit at USD 590.8K (ROI 1.5 $\times$; Figure 3a). The conclusion is robust: the campaign stays profitable in 8 of 9 combinations of success rate ($\pm 50\%$) and cost ($\times 0.5$ – $\times 2$) in the sensitivity analysis.

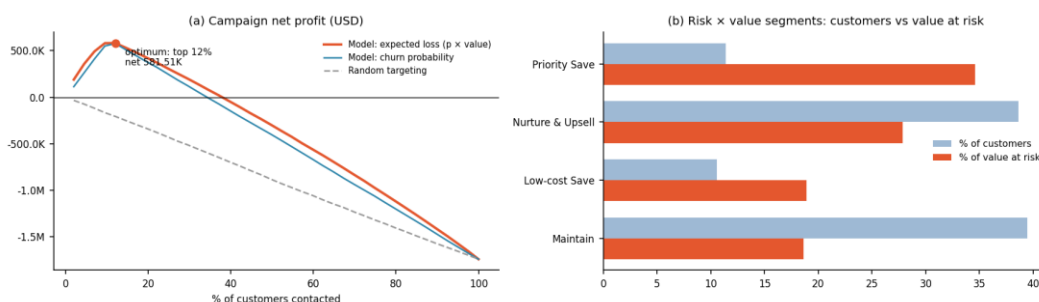


Figure 3. (a) Net profit of a retention campaign by share of customers contacted under three targeting rules; (b) share of customers vs share of value at risk by risk $\times$ value segment.

4.3 Churn personas

Clustering the SHAP explanations of at-risk customers yields 2 personas with different reasons for risk: P2 (79% of at-risk customers, actual churn 8.5%) with moderate, diffuse risk and no dominant driver; P1 (21% of at-risk customers, actual churn 51.9%) driven mainly by days tenure. Each persona receives a different intervention in the recommendations.

5. Recommendations and Implementation

Table 3. Recommended retention actions, ranked by evidence and expected value.

#	Action	Key evidence	Expected impact	KPI	When
1	First-year onboarding programme	'0-6m' churns at 48.8% (4.2x avg); IV 0.89	measured in pilot	13th-month persistency ratio	Medium term
2	Life-stage tailored engagement	'25-34' churns at 14.6% (1.2x avg); IV 0.04	measured in pilot	Churn rate by age band	Medium term
3	Household-based cross-sell and life-event triggers	'0-1' churns at 13.5% (1.2x avg); IV 0.01	measured in pilot	Products per household	Long term
4	Deploy a churn early-warning system	Model ROC-AUC 0.702; top-20% captures 51% of churners	optimal campaign: target top 10% → net 590.81K	Precision/recall of alerts	Quick win
5	Validate with controlled pilots (A/B tests)	What-if results are associational	measured in pilot	Uplift in retention rate (treated – control)	Quick win

1. First-year onboarding programme. Welcome call, policy walkthrough, app activation and 90-day check-in for new customers; first renewal treated as a critical moment. Timing matters: because the churn hazard peaks in tenure interval [90, 180) days, the key touchpoints (welcome call, coverage check, renewal-readiness contact) should be completed before that window opens.

2. Life-stage tailored engagement. Tailor products, pricing communication and service channels to the highest-risk age groups (e.g. digital-first onboarding and flexible payment for younger customers; loyalty recognition for long-standing older customers).

Implementation follows a pilot → measure → scale logic (Figure J.4). Every action keeps a randomised control group: with a baseline churn of 28.0% in the High-risk tier, detecting a 3.5 pp reduction requires 2,481 customers per arm ($\alpha = 0.05$, power 80%). Pilot results then train uplift models so that future campaigns target customers by expected response $\times$ value rather than by risk alone (Ascarza, 2018). The model is deployed as a monthly early-warning score with drift and calibration monitoring and quarterly retraining (Appendix J).

6. Discussion and Conclusion

In answer to the research questions — RQ1: churn is driven mainly by tenure band, age band and length of residence; RQ2: risk peaks in tenure interval [90, 180) days; RQ3: churn can be predicted with ROC-AUC 0.702 (95% CI 0.698–0.707), concentrating 51% of churners in the top 20%; RQ4: a value-targeted campaign returns about USD 590.8K net, while the data contain no controllable levers whose effect can be estimated, so interventions must be tested in pilots. The concentration of exits early in the relationship is consistent with evidence that lapse declines with contract age (Eling & Kiesenbauer, 2014); because acquisition costs are front-loaded, early-tenure retention has the highest value per retained policy.

Two features distinguish this analysis from a standard churn model. First, every claim is supported by converging evidence — univariate tests, multivariate regression, survival models and SHAP — with uncertainty reported as confidence intervals. Second, recommendations are prioritised by money and by causal plausibility, and are designed to be tested. On governance, the score is audited for calibration and selection differences across customer groups and documented in a model card; it is intended for retention outreach only, not for pricing or underwriting (Appendix J).

Limitations. The data are a single snapshot from one insurer, so results may not generalise and may shift with economic conditions. Drivers identified by statistical tests, regression and SHAP are associations; the causal estimates rely on the assumption of no unmeasured confounding, which we probe with E-values but cannot rule out. The campaign economics depend on assumed costs and acceptance rates, which we vary in sensitivity analysis. Variables such as competitor prices, service interactions and household income shocks are not observed. A randomised pilot of the recommended actions is the next step to measure their true effect.

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Appendices

The appendices are not counted in the page limit. They document every analysis step so that each result in the main text can be traced and reproduced; all figures and tables were generated by the accompanying notebook.

Appendix A. Data Overview and Quality

This appendix documents the raw data, data quality checks and the transformations applied before analysis.

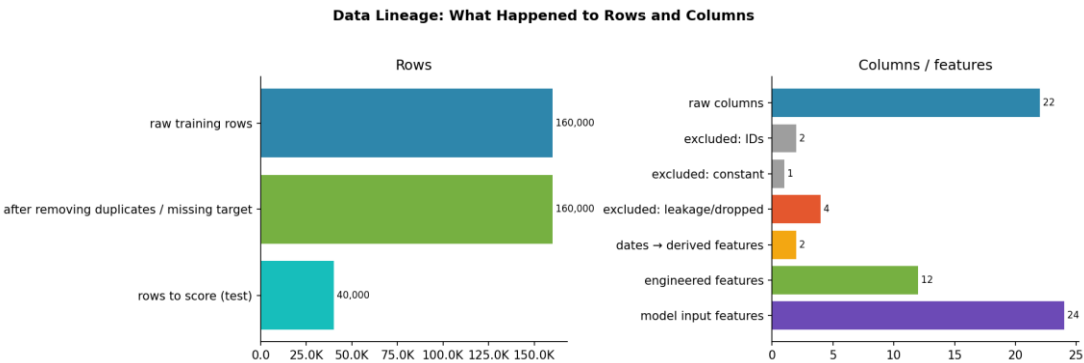


Figure A.1. Data Lineage: What Happened to Rows and Columns

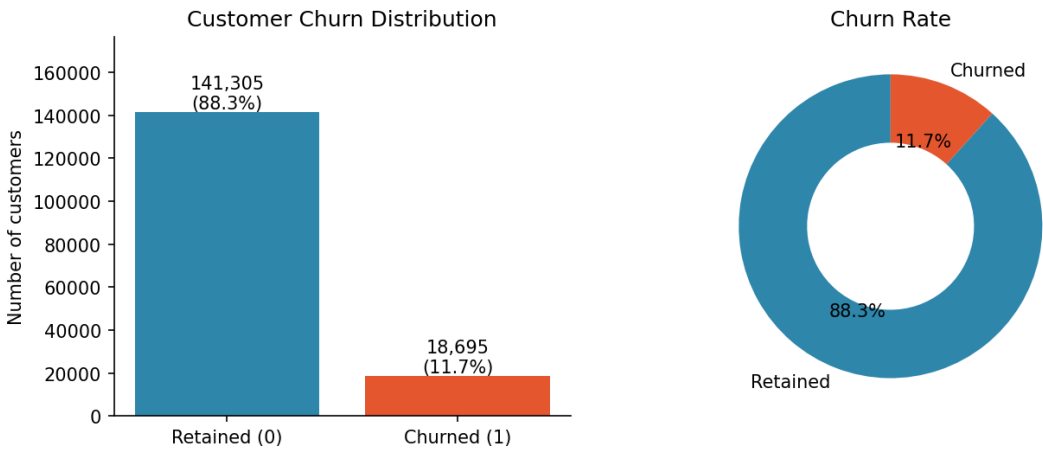


Figure A.2. Customer Churn Distribution

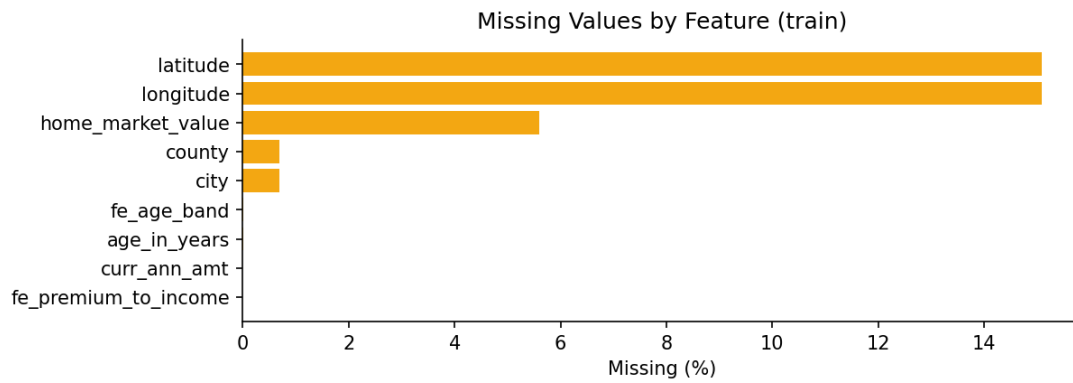


Figure A.3. Missing Values by Feature (train)

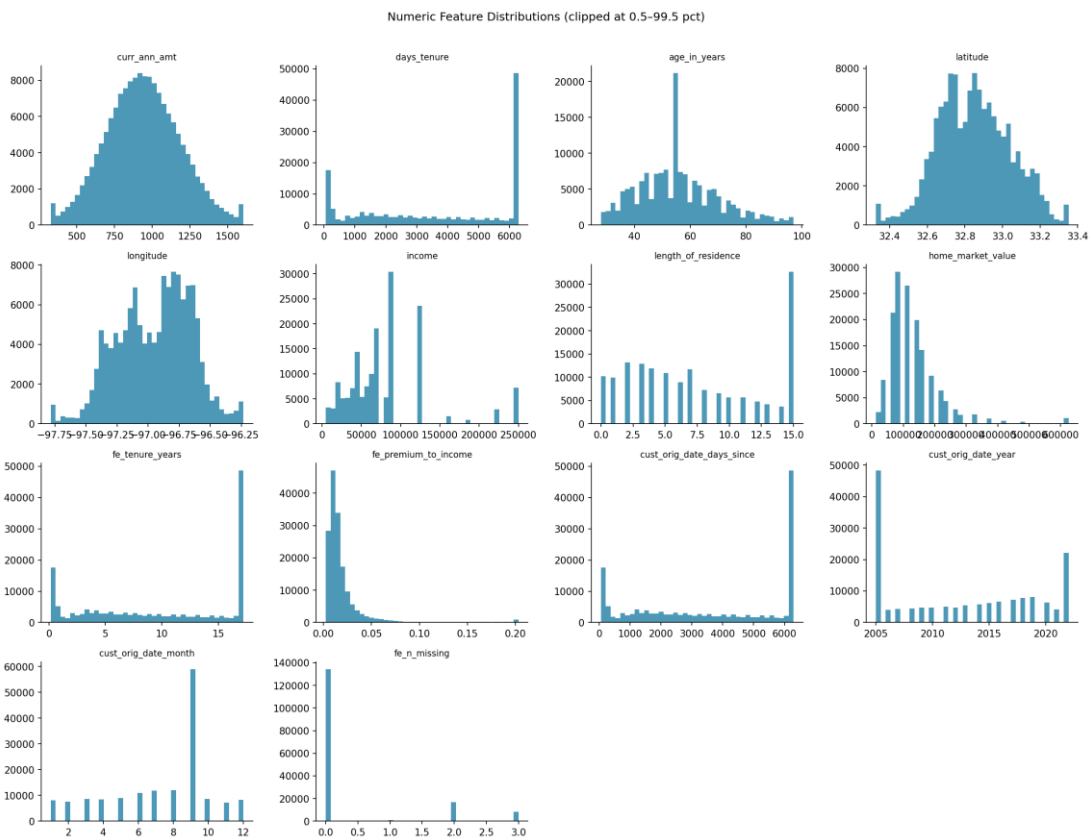


Figure A.4. Numeric Feature Distributions (clipped at 0.5–99.5 pct)

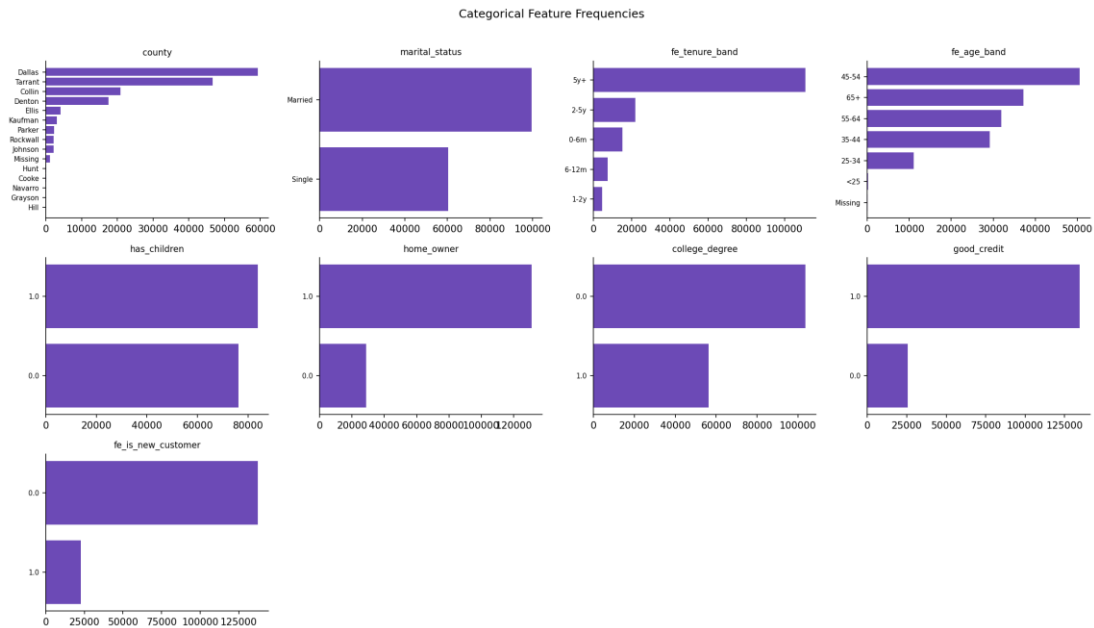


Figure A.5. Categorical Feature Frequencies

Table A.1. Data dictionary and quality report (inferred type, missing values, cardinality).

Column	Inferred type	Role	Missing %	N unique	Top value	Top freq %
individual id	id		0	160,000	221.30B	0
address id	id		0	148,110	521.30B	0.020
curr ann amt	numeric	premium	0	159,979	899.1	0
days tenure	numeric	tenure	0	5,048	6,291	29.5
cust orig date	date		0	5,048	2005-09-11 00:00:00	29.5
age in years	numeric	age	0.010	88	55.0	10.8
date of birth	date		0	1,028	1967-07-07 00:00:00	8.400
latitude	numeric		15.1	124,363	33.0	0.010
longitude	numeric		15.1	124,980	-96.7	0.010
city	high card	region	0.690	95	Dallas	18.3
state	constant		0	1	TX	100.0
county	categorical		0.690	14	Dallas	37.3
income	numeric	income	0	20	87,500	19.0
has children	binary	dependents	0	2	1.000	52.4
length of residence	numeric		0	17	15.0	20.4
marital status	binary_cat		0	2	Married	62.3
home market value	numeric		5.590	18	87,500	19.3
home owner	binary		0	2	1.000	82.0
college degree	binary		0	2	0	64.8
good credit	binary		0	2	1.000	84.0
acct suspd date	date		88.3	300	2022-02-10 00:00:00	0.450

Table A.2. Leakage screen: name-based, missingness-based and single-feature AUC checks.

Feature	Name keyword	Missingness auc	Single feature auc	Statistical evidence	Flagged
acct suspd date year	yes	1.000	1.000	yes	yes
tenure (years)	no		0.696	no	no
days tenure	no		0.696	no	no
cust orig date (days since)	no		0.696	no	no
tenure band	no		0.693	no	no
cust orig date year	no		0.689	no	no
new customer (<12 months)	no		0.680	no	no
age in years	no	0.500	0.547	no	no
age band	no	0.500	0.546	no	no
acct suspd date month	yes	1.000	0.539	yes	yes
length of residence	no		0.530	no	no
cust orig date month	no		0.519	no	no
curr ann amt	no	0.500	0.515	no	no
has children	no		0.515	no	no
college degree	no		0.514	no	no

Automated cleaning log: address_id: kolom ID numerik (nama ID & 183,698 nilai unik) → tidak dipakai sebagai fitur; cust_orig_date: parsed sebagai tanggal (100.0%); date_of_birth: parsed sebagai tanggal (100.0%); home_market_value: string rentang (misal '50000 - 74999') → numeric titik tengah; acct_suspd_date: parsed sebagai tanggal (100.0%)

Appendix B. Methodology and Validation Design

Architecture of the analytics pipeline, the cross-validation design and the predictive model, plus tuning results.

Statistical screening. For categorical variables we use Pearson's χ^2 test with Cramér's V; for numeric variables the Mann–Whitney U test with the rank-biserial correlation, so that every p-value is paired with an effect size. Because dozens of variables are tested, p-values are converted to Benjamini–Hochberg q-values. Weight of evidence and Information Value, standard in credit and insurance scoring, summarise each binned variable's predictive strength (Siddiqi, 2006); proportions are reported with Wilson 95% intervals.

Survival layer. Tenure is the time axis and active customers are right-censored. Kaplan–Meier curves and log-rank tests describe retention; discrete hazards per tenure interval locate the critical window; the Cox model gives adjusted hazard ratios, its proportional-

hazards assumption is checked with Schoenfeld residuals (Schoenfeld, 1982), and a Weibull accelerated failure time model provides time ratios that remain interpretable when the assumption fails (Wei, 1992). Restricted mean survival time expresses differences in months or days retained (Royston & Parmar, 2013).

Predictive layer. Logistic regression, random forest, extra trees, histogram gradient boosting, LightGBM, XGBoost and CatBoost (Ke et al., 2017; Chen & Guestrin, 2016; Prokhorenkova et al., 2018), a glass-box Explainable Boosting Machine (Lou et al., 2013) and a neural network are trained on identical folds. Hyperparameters are tuned with the TPE sampler and median pruning (Akiba et al., 2019) on inner folds; the best families are re-trained with several seeds; features with non-positive permutation importance are tested for removal; and six ensembling strategies, including greedy ensemble selection (Caruana et al., 2004) and stacking, are compared, the simplest within 0.0005 of the best being preferred. Probabilities are calibrated by isotonic or Platt scaling when this lowers cross-validated log-loss, and calibration is summarised by the Brier score and expected calibration error (Guo et al., 2017).

Model comparison. Correlated AUCs are compared with the DeLong test (DeLong et al., 1988) using the fast algorithm, with Holm correction across comparisons; bootstrap intervals complement the analytic ones. Because AUC is not aligned with campaign profit (Verbeke et al., 2012; Höppner et al., 2020), models are also ranked by the expected maximum profit measure for churn, which integrates campaign profit over a Beta-distributed acceptance rate (Verbraken et al., 2013). The risk score is further validated on time-to-churn with Harrell's C-index (Harrell et al., 1982).

Why not only an interpretable model? Interpretable models should be preferred when they perform comparably (Rudin, 2019); we therefore report the logistic regression and the EBM next to the ensemble and test whether the accuracy gain is significant.

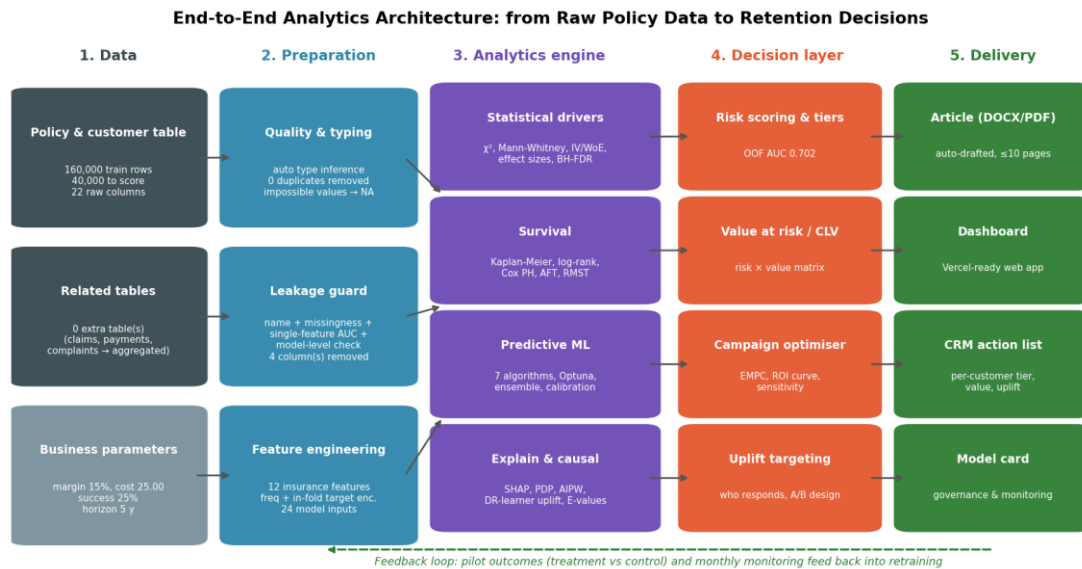


Figure B.1. End-to-end analytics architecture: from raw policy data to retention decisions, with a feedback loop from pilots.

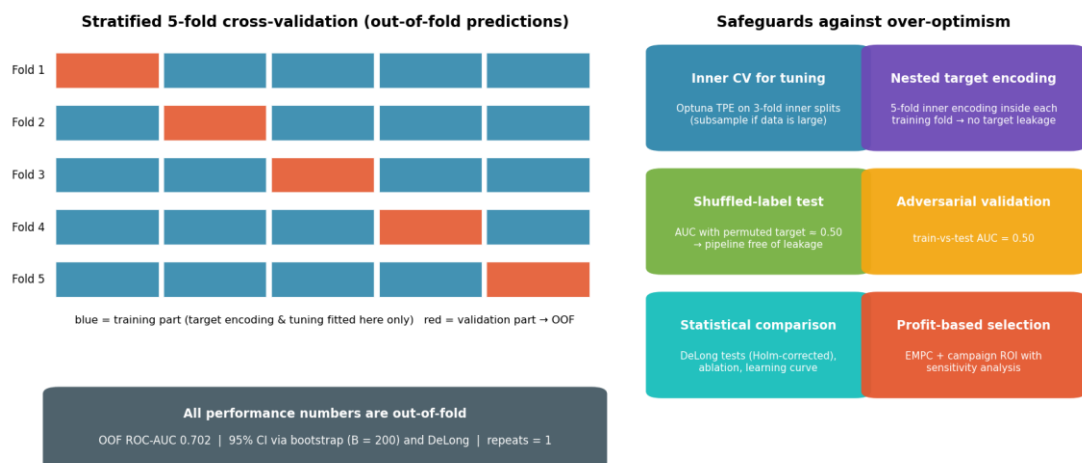


Figure B.2. validation_design

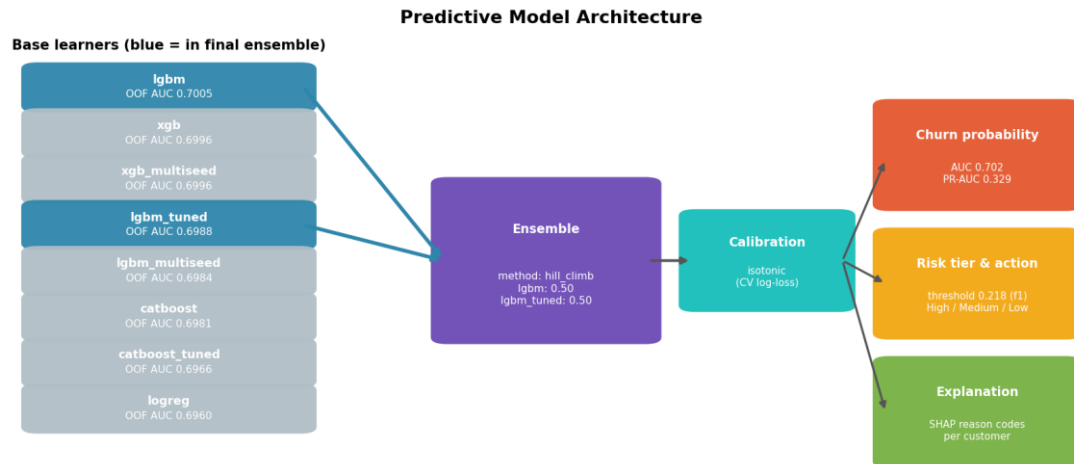


Figure B.3. Predictive Model Architecture

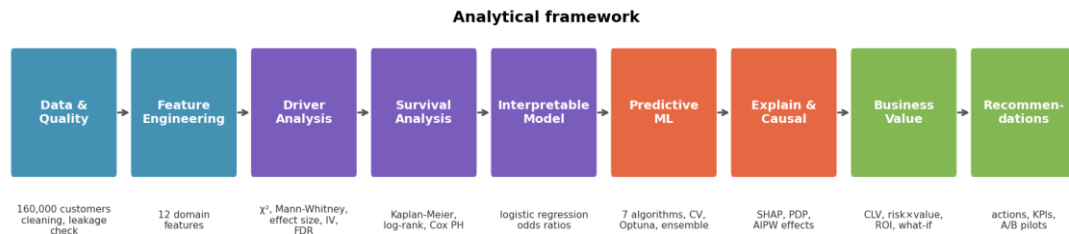


Figure B.4. Analytical framework

Table B.1. Key analysis settings (reproducibility).

Setting	Value
RUN MODE	full
SEED	42
N_FOLDS	5
N_REPEATS	1
METRIC	roc_auc
MODELS	['logreg', 'hgb', 'lgbm', 'xgb', 'catboost', 'ebm', 'mlp']
USE_OPTUNA	True
OPTUNA_TRIALS	15
FINAL_SEEDS	[42, 7]
CALIBRATION	auto
PROFIT_MARGIN	0.15
RETENTION_COST	25
RETENTION_SUCCESS_RATE	0.25
DISCOUNT_RATE	0.1
CLV_HORIZON_YEARS	5
RISK_TIER_CUTS	(0.5, 0.8)
WHAT_IF_ADOPTION	0.3
PILOT_POWER	0.8
PILOT_ALPHA	0.05

Table B.2. Bayesian hyperparameter optimisation (Optuna) results.

Model	Trials completed	Best tuning score	Default cv auc	Best params	Tuned cv auc	Gain auc
lgbm	11	0.703	0.700	{ "learning_rate": 0.07696668733291795, "num_leaves": 130, "min_child_samples": 11, "subsample": 0.512121241850058, "colsample_bytree": 0.6036841581615148, "reg_alpha": 0.0195536152465732, "reg_lambda": 6.703489882056627e-06, "min_split_gain": 0.03358588493447223, "cat_smooth": 36.61527549847105 }	0.699	-0.002
catboost	6	0.699	0.698	{ "learning_rate": 0.09729870597990298, "depth": 4, "l2_leaf_reg": 5.748924681991978, "random_strength": 1.7772437065861273, "bagging_temperature": 0.046450412719997725 }	0.697	-0.001
xgb	11	0.702	0.700	{ "learning_rate": 0.08813259224734658, "max_depth": 5, "min_child_weight": 4.231156845697541, "subsample": 0.5089809378074099, "colsample_bytree": 0.6110461787176937, "reg_alpha": 0.0015027010896820222, "reg_lambda": 4.3444691085504035, "gamma": 3.3828618987138985 }	0.696	-0.004

Appendix C. Driver Analysis (Detailed)

Churn rates for every analysed variable with 95% confidence intervals, predictive strength (Information Value), correlations and interactions.

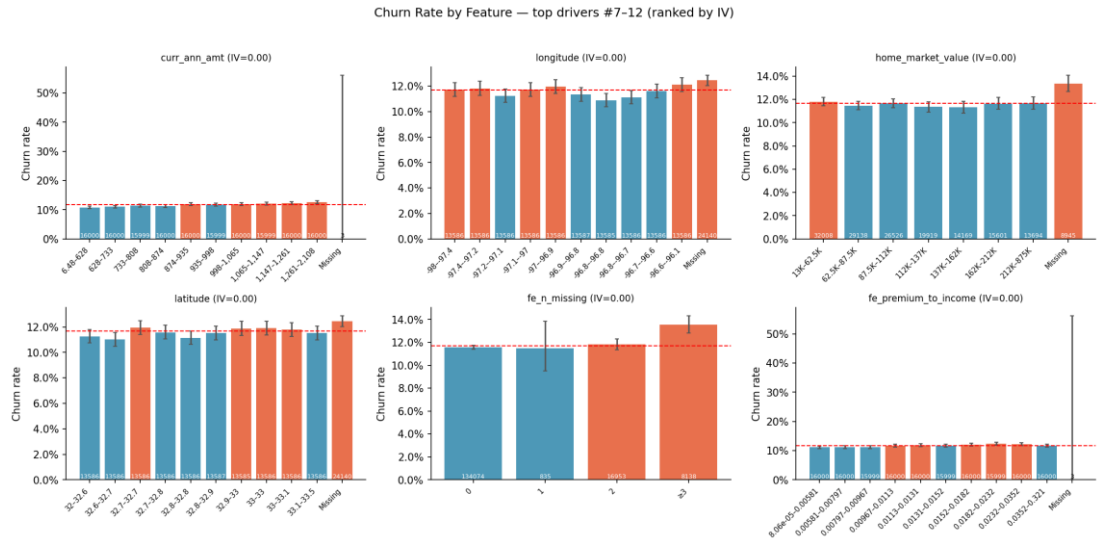


Figure C.1. Churn Rate by Feature — top drivers #7–12 (ranked by IV)

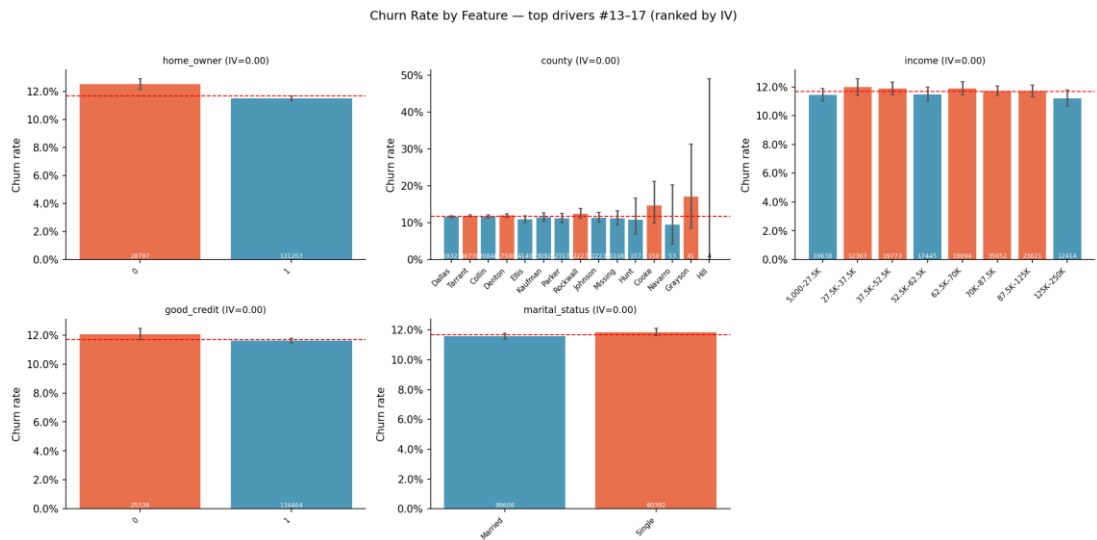


Figure C.2. Churn Rate by Feature — top drivers #13–17 (ranked by IV)

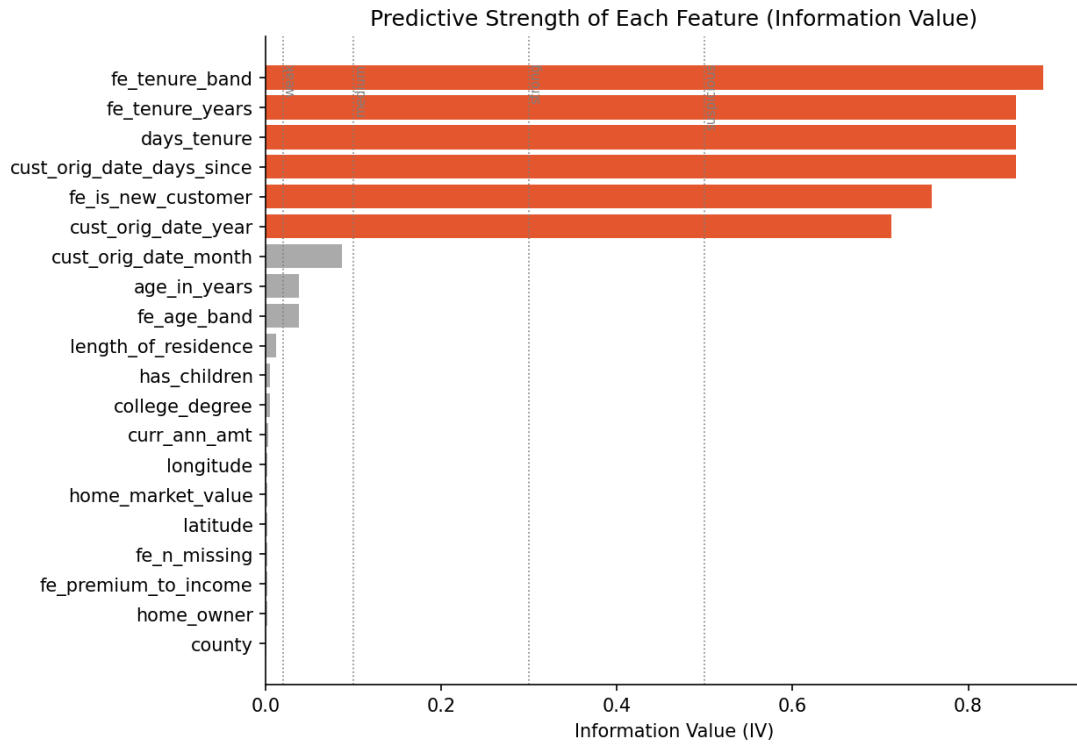


Figure C.3. Predictive Strength of Each Feature (Information Value)

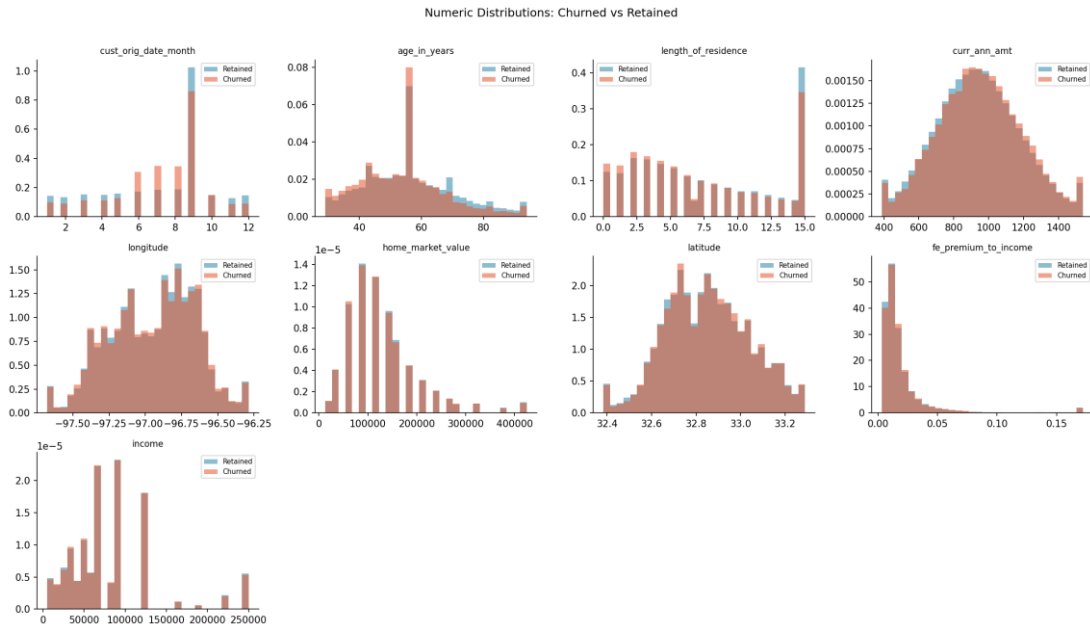


Figure C.4. Numeric Distributions: Churned vs Retained

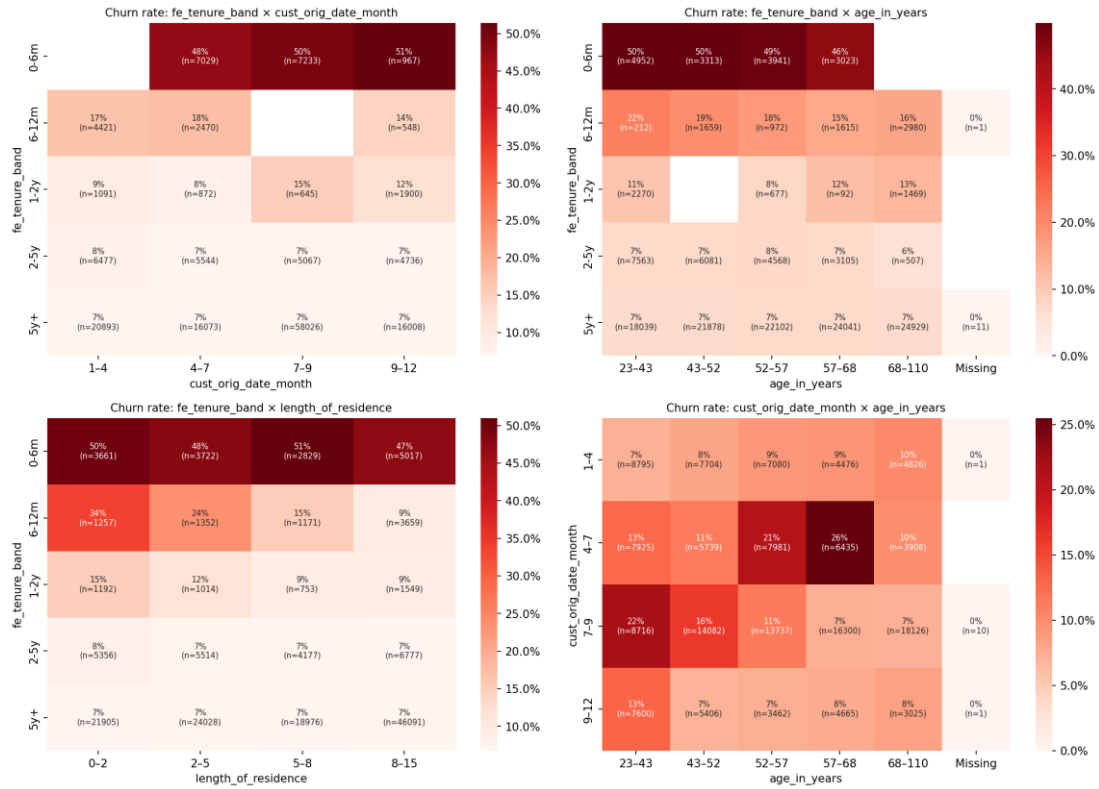


Figure C.5. Churn rate: tenure band $\times$ cust orig date month

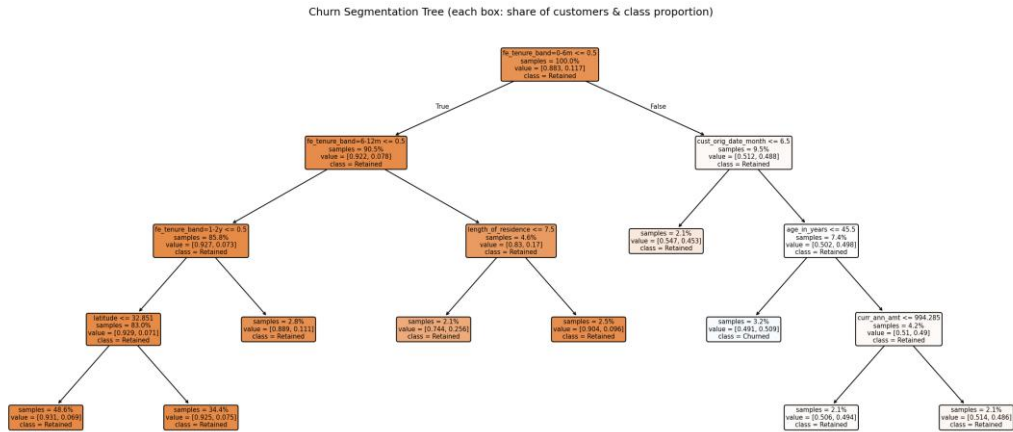


Figure C.6. Churn Segmentation Tree (each box: share of customers & class proportion)

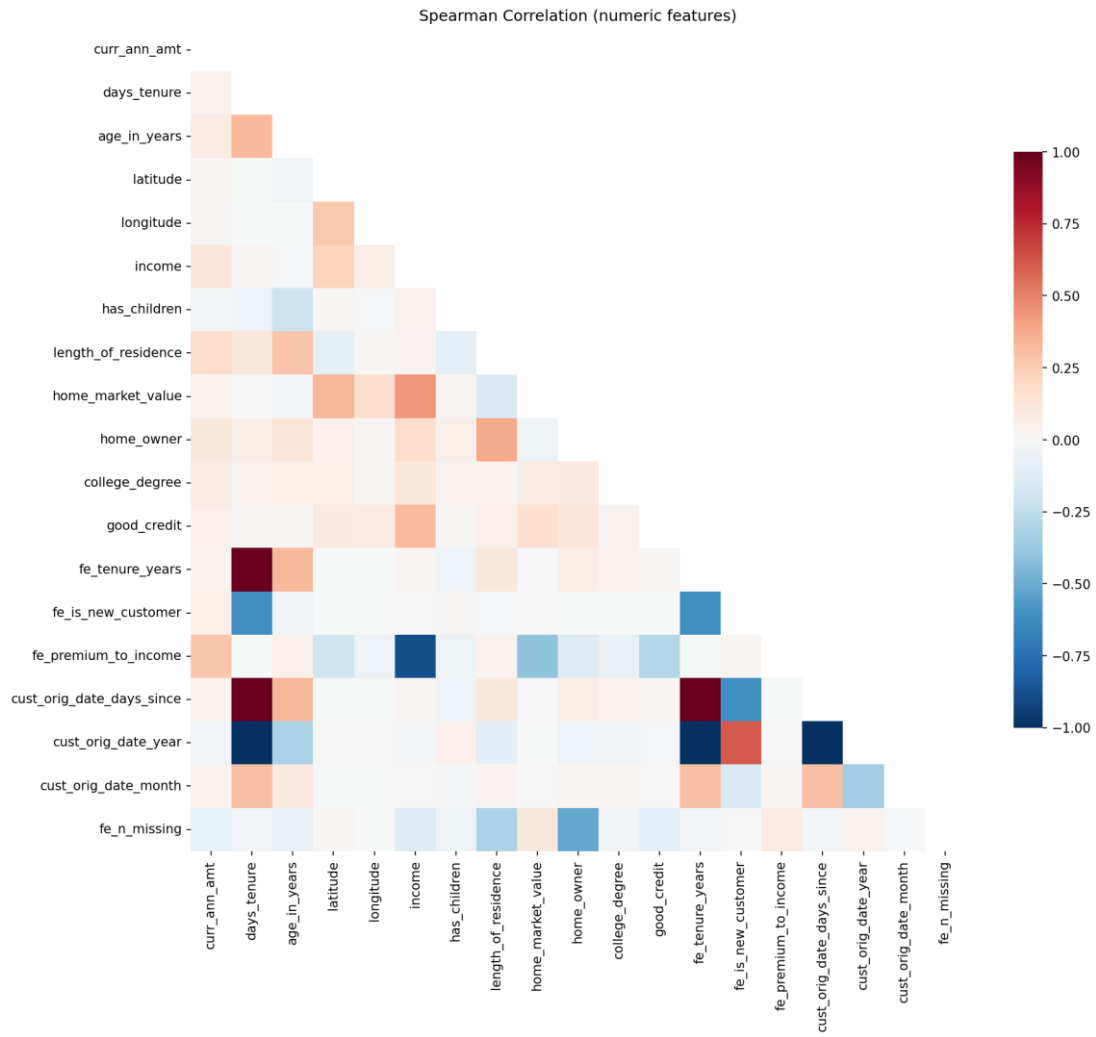


Figure C.7. Spearman Correlation (numeric features)

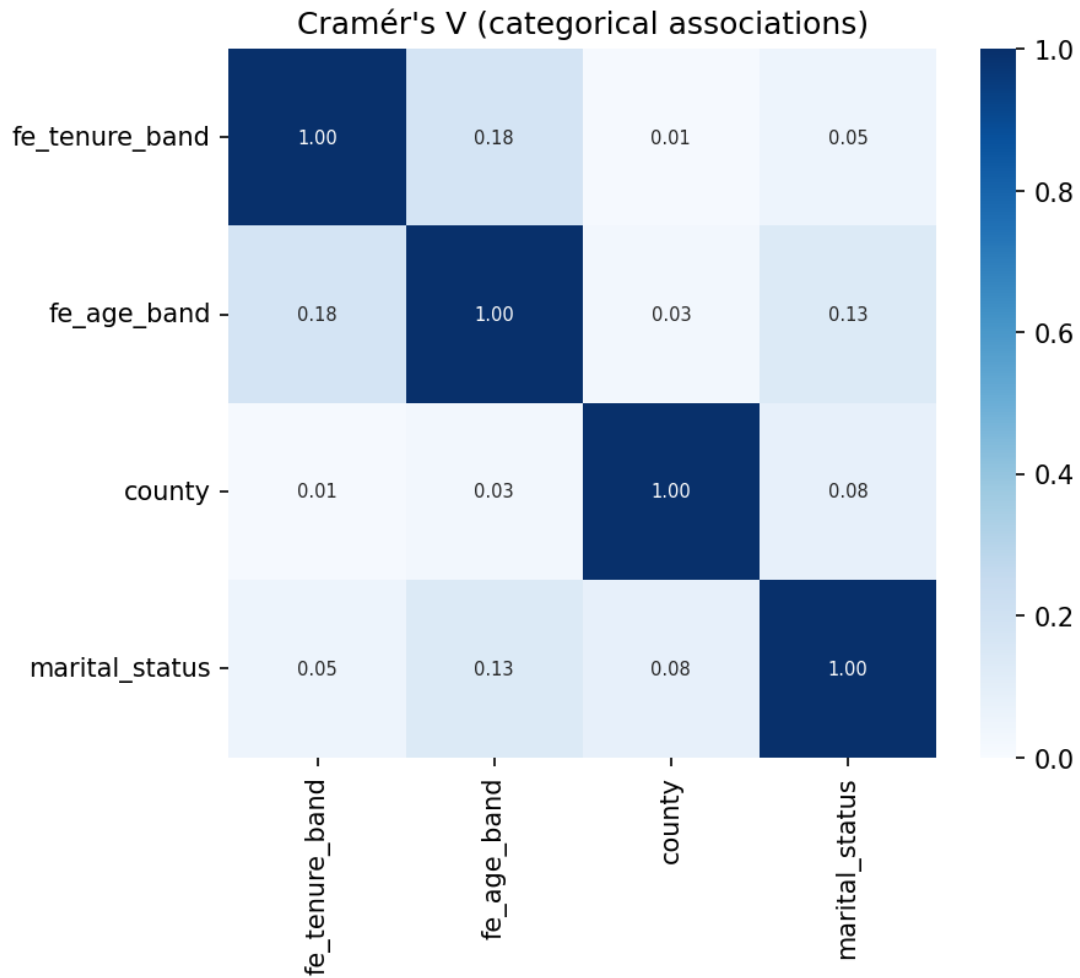


Figure C.8. Cramér's V (categorical associations)

Table C.1. Univariate driver statistics (top 25 by Information Value; q-values Benjamini–Hochberg).

Feature	Type	Iv	Iv strength	Effect metric	Effect size	Q value BH	Univariate auc	Highest risk level	Highest risk rate
tenure band	categorical	0.885	suspicious(>0.5)	Cramér's V	0.381	0	0.693	0–6m	0.488
tenure (years)	numeric	0.855	suspicious(>0.5)	rank-biserial r	-0.391	0	0.696	0.0548–0.523	0.473
days tenure	numeric	0.855	suspicious(>0.5)	rank-biserial r	-0.391	0	0.696	20–191	0.473
cust orig date (days since)	numeric	0.855	suspicious(>0.5)	rank-biserial r	-0.391	0	0.696	20–191	0.473
new customer (<12 months)	categorical	0.759	suspicious(>0.5)	Cramér's V	0.338	0	0.680	1	0.384
cust orig date year	numeric	0.713	suspicious(>0.5)	rank-biserial r	0.378	0	0.689	2,020–2,022	0.348
cust orig date month	numeric	0.086	weak	rank-biserial r	-0.039	1.2e-18	0.519	6–7	0.201
age in years	numeric	0.038	weak	rank-biserial r	-0.094	3.8e-96	0.547	23–37	0.143

Feature	Type	Iv	Iv strength	Effect metric	Effect size	Q value BH	Univariate auc	Highest risk level	Highest risk rate
age band	categorical	0.038	weak	Cramér's V	0.060	3.1e-120	0.546	25-34	0.146
length of residence	numeric	0.011	useless	rank-biserial r	-0.061	1.9e-41	0.530	0-1	0.135
has children	categorical	0.005	useless	Cramér's V	0.022	1.2e-17	0.515	1	0.123
college degree	categorical	0.004	useless	Cramér's V	0.021	5.4e-17	0.514	0	0.122
curr ann amt	numeric	0.003	useless	rank-biserial r	0.030	3.8e-11	0.515	1,261-2,108	0.126
longitude	numeric	0.002	useless	rank-biserial r	-0.001	0.772	0.501	Missing	0.124
home market value	numeric	0.002	useless	rank-biserial r	-0.004	0.433	0.502	Missing	0.134
latitude	numeric	0.002	useless	rank-biserial r	0.009	0.082	0.505	Missing	0.124
number of missing fields	numeric	0.002	useless	rank-biserial r	0.011	1.0e-04	0.506	≥3	0.136
premium-to-income ratio	numeric	0.002	useless	rank-biserial r	0.017	1.7e-04	0.509	0.0182-0.0232	0.124
home owner	categorical	0.001	useless	Cramér's V	0.012	3.4e-06	0.504	0	0.125
county	categorical	6.3e-04	useless	Cramér's V	0.008	0.724	0.500	Rockwall	0.124
income	numeric	4.3e-04	useless	rank-biserial r	-0.003	0.624	0.501	27.5K-37.5K	0.120
good credit	categorical	2.6e-04	useless	Cramér's V	0.005	0.052	0.502	0	0.121
marital status	categorical	1.7e-04	useless	Cramér's V	0.004	0.113	0.502	Single	0.119

Table C.2. All decision-tree segments.

Segment rule	N customers	Share %	Churn rate	Lift	Churners captured %
tenure band = 0-6m AND cust orig date month > 6.50 AND age in years ≤ 45.50	5,105	3.191	0.509	4.359	13.9
tenure band = 0-6m AND cust orig date month > 6.50 AND age in years > 45.50 AND curr ann amt ≤ 994.28	3,404	2.127	0.494	4.226	8.992
tenure band = 0-6m AND cust orig date month > 6.50 AND age in years > 45.50 AND curr ann amt > 994.28	3,367	2.104	0.486	4.161	8.756
tenure band = 0-6m AND cust orig date month ≤ 6.50	3,353	2.096	0.453	3.880	8.131
tenure band ≠ 0-	3,425	2.141	0.256	2.191	4.691

Segment rule	N customers	Share %	Churn rate	Lift	Churners captured %
6m AND tenure band = 6-12m AND length of residence ≤ 7.50					
tenure band $\neq$ 0-6m AND tenure band $\neq$ 6-12m AND tenure band = 1-2y	4,508	2.817	0.111	0.953	2.685
tenure band $\neq$ 0-6m AND tenure band = 6-12m AND length of residence > 7.50	4,014	2.509	0.096	0.819	2.054
tenure band $\neq$ 0-6m AND tenure band $\neq$ 6-12m AND tenure band $\neq$ 1-2y AND latitude > 32.85	55,107	34.4	0.075	0.639	22.0
tenure band $\neq$ 0-6m AND tenure band $\neq$ 6-12m AND tenure band $\neq$ 1-2y AND latitude ≤ 32.85	77,717	48.6	0.069	0.593	28.8

Table C.3. Variance inflation factors (numeric features).

Feature	VIF
days tenure	1,000,799.92B
tenure (years)	1,000,799.92B
cust orig date (days since)	1,000,799.92B
cust orig date year	103,605
cust orig date month	147.2
new customer (<12 months)	1.621
home owner	1.564
income	1.517
premium-to-income ratio	1.489
number of missing fields	1.469
length of residence	1.329
age in years	1.271
good credit	1.188
home market value	1.175
curr ann amt	1.112
latitude	1.109
has children	1.065
longitude	1.057
college degree	1.034

Appendix D. Interpretable Logistic Regression

Adjusted odds ratios with 95% confidence intervals; numeric variables are standardised (per 1 SD), categories are compared with the most common level.

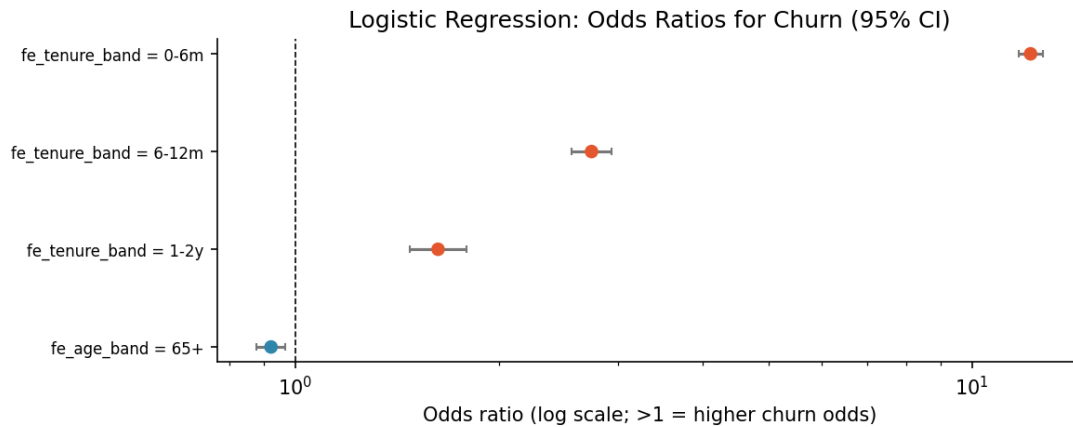


Figure D.1. Logistic Regression: Odds Ratios for Churn (95% CI)

Table D.1. Logistic regression: odds ratios, 95% CI, p-values and average marginal effects.

Term	Coef	Odds ratio	Or ci low	Or ci high	P value	Scale	Avg marginal effect pp	Interpretation
age band = Missing	-16.8	5.2e-08	0	infB	0.997	vs '45-54'	-148.0	-100% odds of churn (vs '45-54')
tenure band = 0-6m	2.502	12.2	11.7	12.7	0	vs '5y+'	22.1	+1120% odds of churn (vs '5y+')
tenure band = 6-12m	1.008	2.740	2.558	2.936	1.5e-180	vs '5y+'	8.892	+174% odds of churn (vs '5y+')
tenure band = 1-2y	0.486	1.626	1.475	1.792	1.5e-22	vs '5y+'	4.287	+63% odds of churn (vs '5y+')
age band = <25	-0.301	0.740	0.441	1.242	0.255	vs '45-54'	-2.653	-26% odds of churn (vs '45-54')
age band = 65+	-0.082	0.921	0.877	0.968	0.001	vs '45-54'	-0.725	-8% odds of churn (vs '45-54')
age band = 25-34	0.047	1.048	0.981	1.120	0.163	vs '45-54'	0.417	+5% odds of churn (vs '45-54')
age band = 35-44	-0.015	0.985	0.940	1.033	0.544	vs '45-54'	-0.130	-1% odds of churn (vs '45-54')
cust orig date month	0.011	1.011	0.992	1.030	0.250	per 1 SD (= 2.93)	0.097	+1% odds of churn (per 1 SD (= 2.93))
tenure band = 2-5y	0.010	1.010	0.953	1.069	0.741	vs '5y+'	0.085	+1% odds of churn (vs '5y+')
age band = 55-64	5.5e-04	1.001	0.955	1.048	0.982	vs '45-54'	0.005	+0% odds of churn (vs '45-54')

Appendix E. Survival Analysis

Kaplan–Meier estimates, hazards by tenure, Cox and Weibull AFT models, proportional-hazards diagnostics and validation of risk tiers over time.

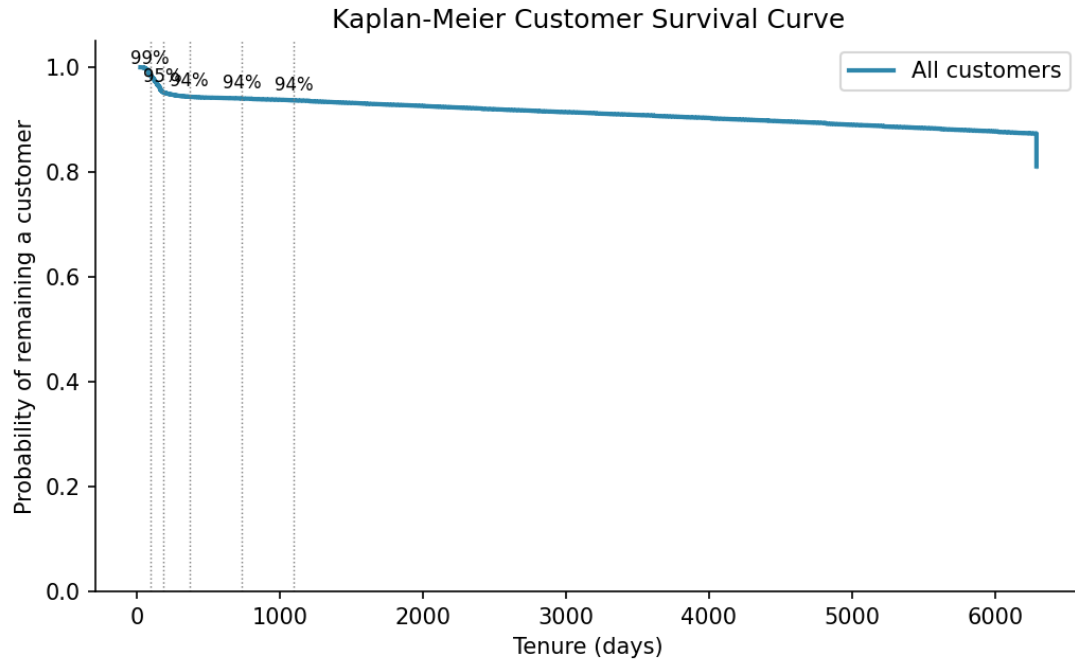


Figure E.1. Kaplan-Meier Customer Survival Curve

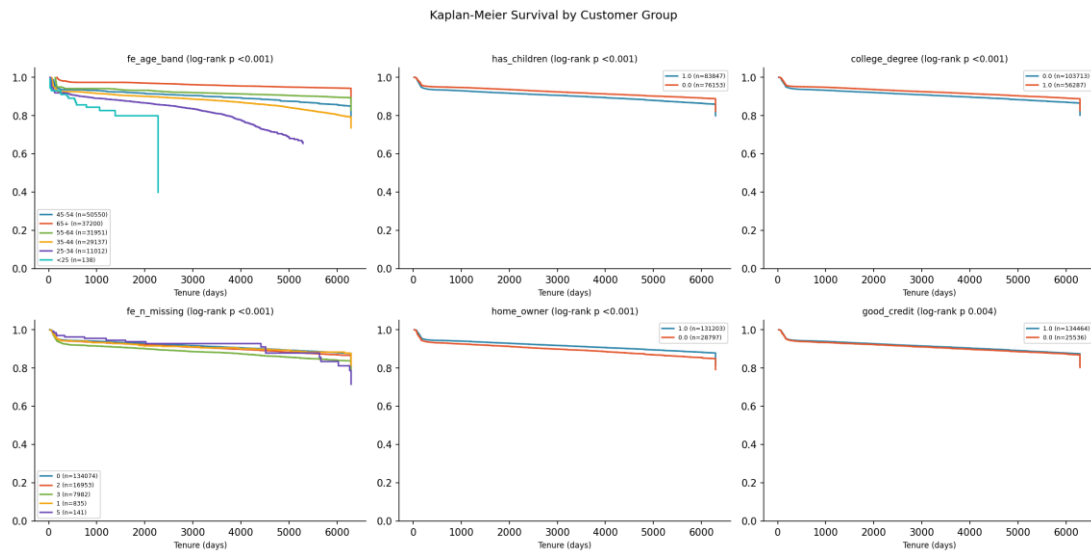


Figure E.2. Kaplan-Meier Survival by Customer Group

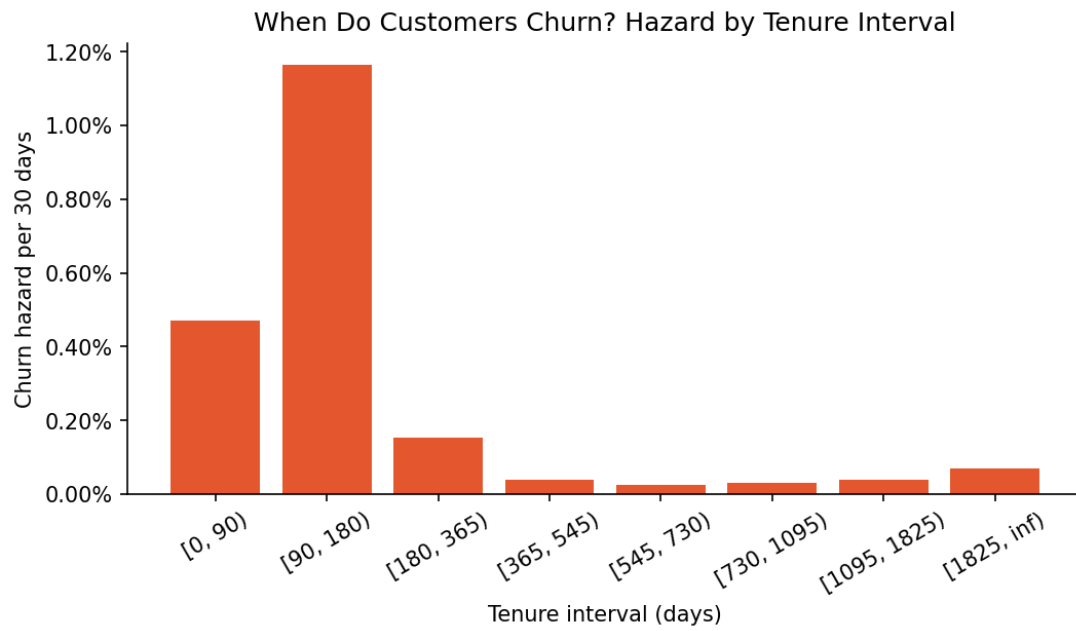


Figure E.3. When Do Customers Churn? Hazard by Tenure Interval

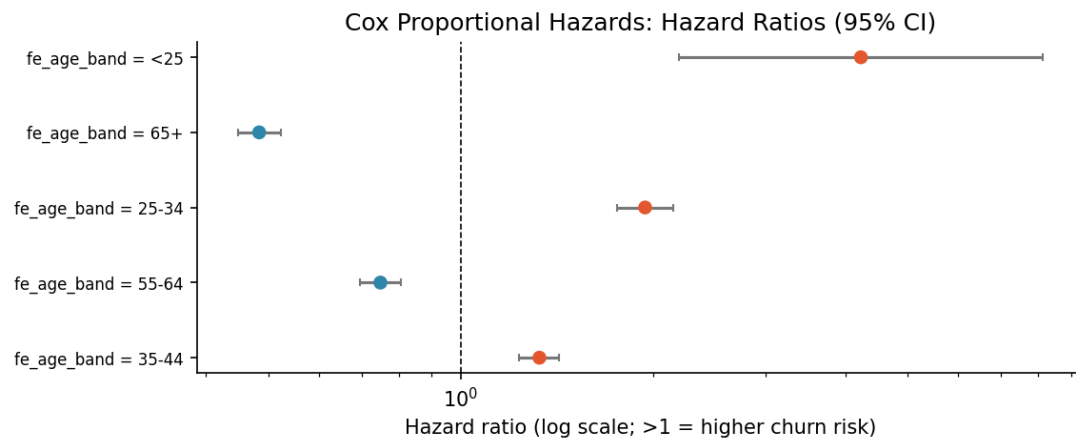


Figure E.4. Cox Proportional Hazards: Hazard Ratios (95% CI)

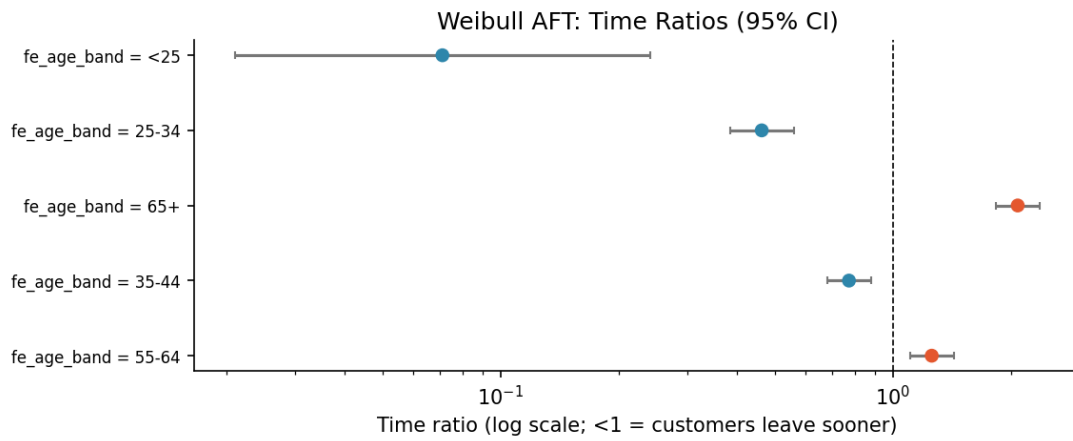


Figure E.5. Weibull AFT: Time Ratios (95% CI)

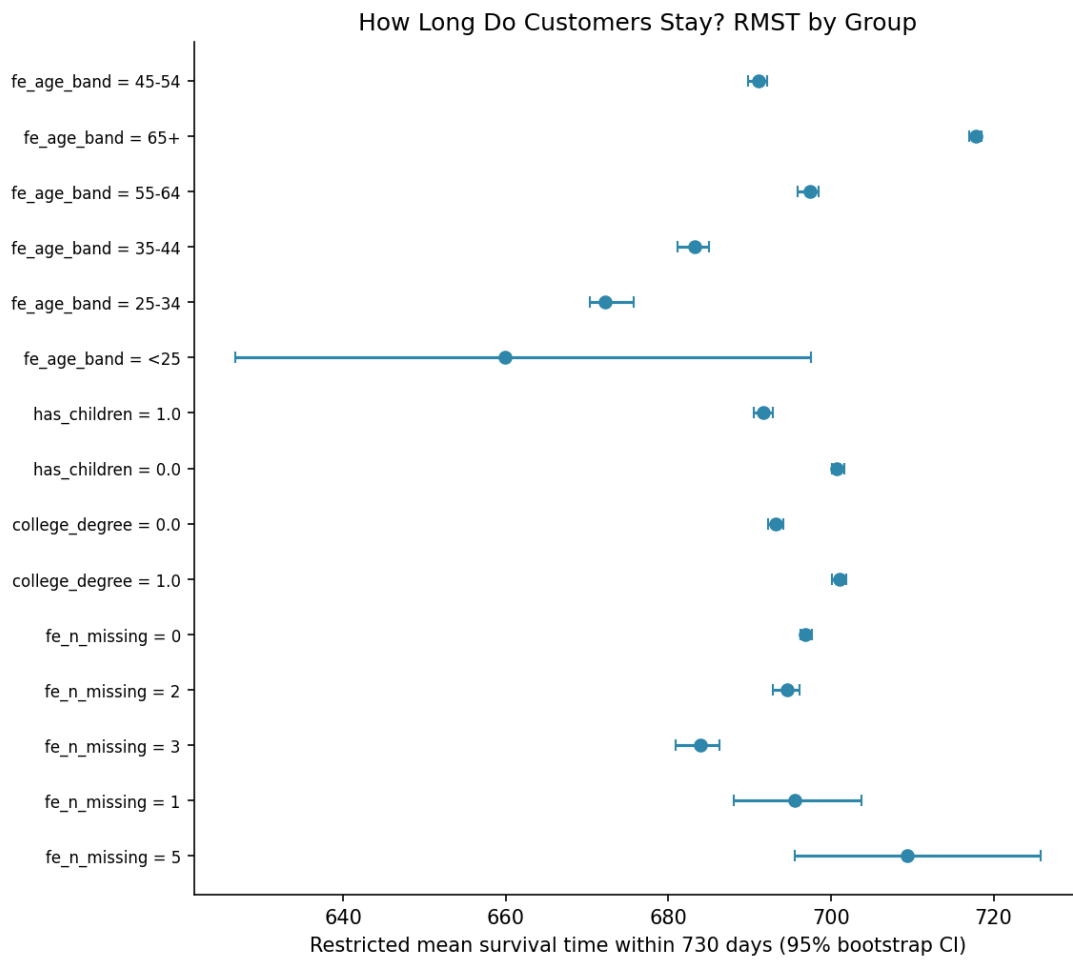


Figure E.6. How Long Do Customers Stay? RMST by Group

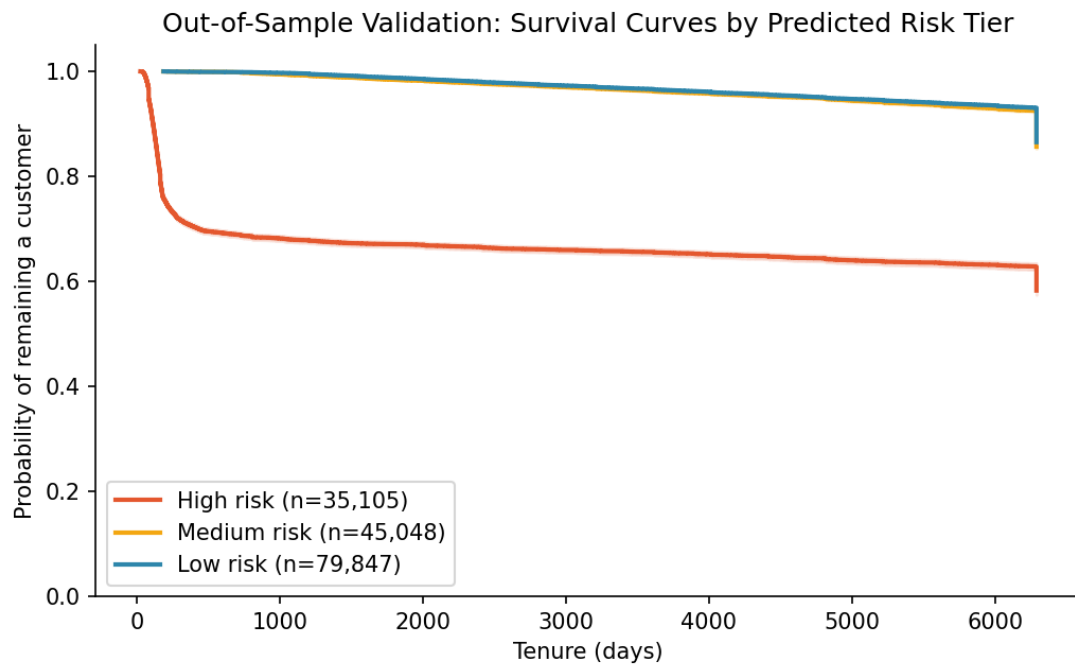


Figure E.7. Out-of-Sample Validation: Survival Curves by Predicted Risk Tier

Table E.1. Hazard by tenure interval.

Interval	At risk start	Churn events	Person time	Hazard
[0, 90)	160,000	2,215	14.32M	0.005
[90, 180)	155,521	5,176	13.53M	0.012
[180, 365)	145,025	1,308	25.92M	0.002
[365, 545)	137,350	305	24.51M	3.8e-04
[545, 730)	135,208	196	24.85M	2.4e-04
[730, 1095)	132,836	475	47.41M	3.0e-04
[1095, 1825)	126,907	1,116	86.62M	3.9e-04
[1825, inf)	111,008	7,904	338.55M	7.1e-04

Table E.2. Log-rank tests by group.

Feature	Logrank chi2	P value
age band	2,877	0
has children	174.0	0
college degree	132.6	0
number of missing fields	76.0	1.2e-15
home owner	101.2	0
good credit	8.325	0.004

Table E.3. Cox proportional hazards model.

Term	Coef	Hazard ratio	Hr ci low	Hr ci high	P value	Scale	Interpretation
age band = Missing	-16.2	9.0e-08	0	infB	0.997	vs '45-54'	-100% churn hazard (vs '45-54')
age band = <25	1.441	4.225	2.194	8.137	1.6e-05	vs '45-54'	+323% churn hazard (vs '45-54')
age band =	-0.725	0.484	0.448	0.523	3.7e-76	vs '45-54'	-52% churn

Term	Coef	Hazard ratio	Hr ci low	Hr ci high	P value	Scale	Interpretation
65+							hazard (vs '45-54')
age band = 25-34	0.664	1.943	1.756	2.149	5.9e-38	vs '45-54'	+94% churn hazard (vs '45-54')
age band = 55-64	-0.289	0.749	0.697	0.806	5.7e-15	vs '45-54'	-25% churn hazard (vs '45-54')
age band = 35-44	0.284	1.328	1.236	1.427	1.2e-14	vs '45-54'	+33% churn hazard (vs '45-54')

Table E.4. Schoenfeld test of the proportional-hazards assumption.

Term	Test statistic	P	Ph violated
age band = <25	0.826	0.363	no
age band = 25-34	0.657	0.418	no
age band = 35-44	9.621	0.002	yes
age band = 55-64	5.314	0.021	no
age band = 65+	80.4	3.1e-19	yes
age band = Missing	0.002	0.966	no

Table E.5. Weibull accelerated failure time model (time ratios).

Term	Time ratio	Tr ci low	Tr ci high	P value	Scale
age band = <25	0.071	0.021	0.241	2.1e-05	vs '45-54'
age band = Missing	11.7	0.002	70,299	0.580	vs '45-54'
age band = 25-34	0.463	0.384	0.559	7.9e-16	vs '45-54'
age band = 65+	2.082	1.828	2.371	2.5e-28	vs '45-54'
age band = 35-44	0.773	0.679	0.880	1.0e-04	vs '45-54'
age band = 55-64	1.257	1.106	1.429	4.7e-04	vs '45-54'

Table E.6. Restricted mean survival time by group.

Feature	Level	N	Rmst	Rmst ci low	Rmst ci high	Diff vs best	Reference level	Tau
age band	45-54	50,550	691.0	689.8	692.2	-26.8	65+	730.0
age band	65+	37,200	717.9	717.0	718.5	0	65+	730.0
age band	55-64	31,951	697.4	695.9	698.5	-20.4	65+	730.0
age band	35-44	29,137	683.2	681.1	685.0	-34.6	65+	730.0
age band	25-34	11,012	672.2	670.4	675.7	-45.6	65+	730.0
age band	<25	138	659.9	626.7	697.6	-57.9	65+	730.0
has children	1.0	83,847	691.7	690.5	692.8	-9.058	0.0	730.0
has children	0.0	76,153	700.7	700.2	701.6	0	0.0	730.0
college degree	0.0	103,713	693.2	692.3	694.1	-7.844	1.0	730.0
college degree	1.0	56,287	701.1	700.1	701.9	0	1.0	730.0
number of missing fields	0	134,074	696.9	696.2	697.6	-12.5	5	730.0
number of missing fields	2	16,953	694.6	692.9	696.1	-14.7	5	730.0
number of missing fields	3	7,982	684.0	680.9	686.3	-25.4	5	730.0
number of missing fields	1	835	695.6	688.1	703.8	-13.8	5	730.0

Feature	Level	N	Rmst	Rmst ci low	Rmst ci high	Diff vs best	Reference level	Tau
number of missing fields	5	141	709.3	695.6	725.8	0	5	730.0

Table E.7. Survival probability by predicted risk tier.

Tier	N	S(90)	S(180)	S(365)	S(730)	S(1095)
High	35,105	0.933	0.760	0.708	0.687	0.679
Medium	45,048	1.000	1.000	0.999	0.998	0.993
Low	79,847	1.000	1.000	0.999	0.999	0.997

Appendix F. Model Development, Comparison and Validation

All logged experiments (trial-and-error), statistical comparison of models, ablation study and diagnostic checks.

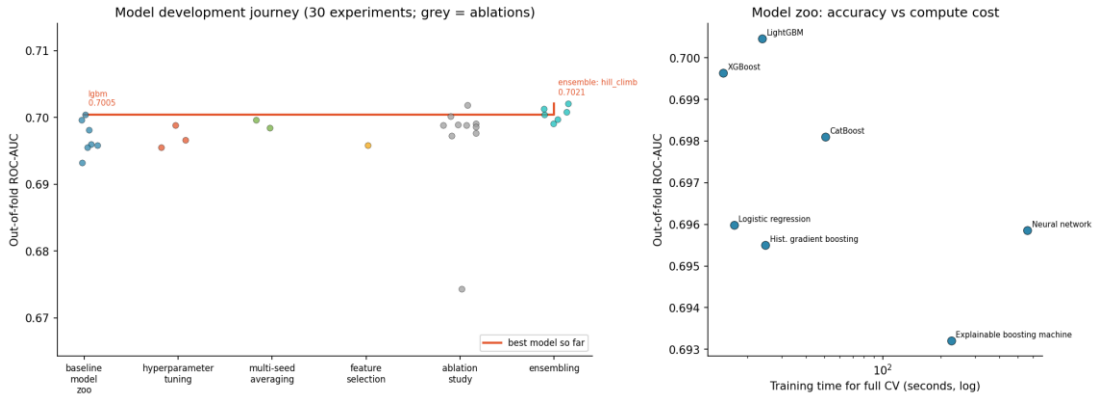


Figure F.1. Model development journey (30 experiments; grey = ablations)

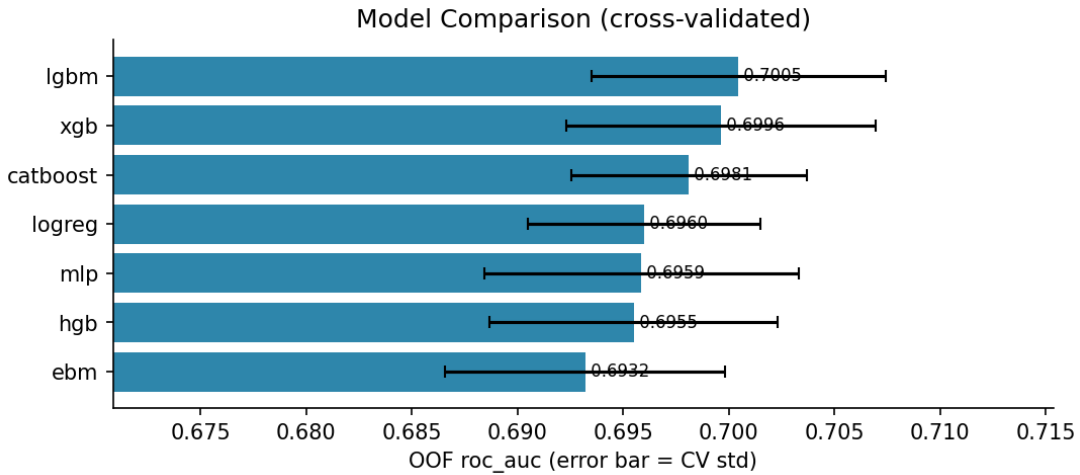


Figure F.2. Model Comparison (cross-validated)

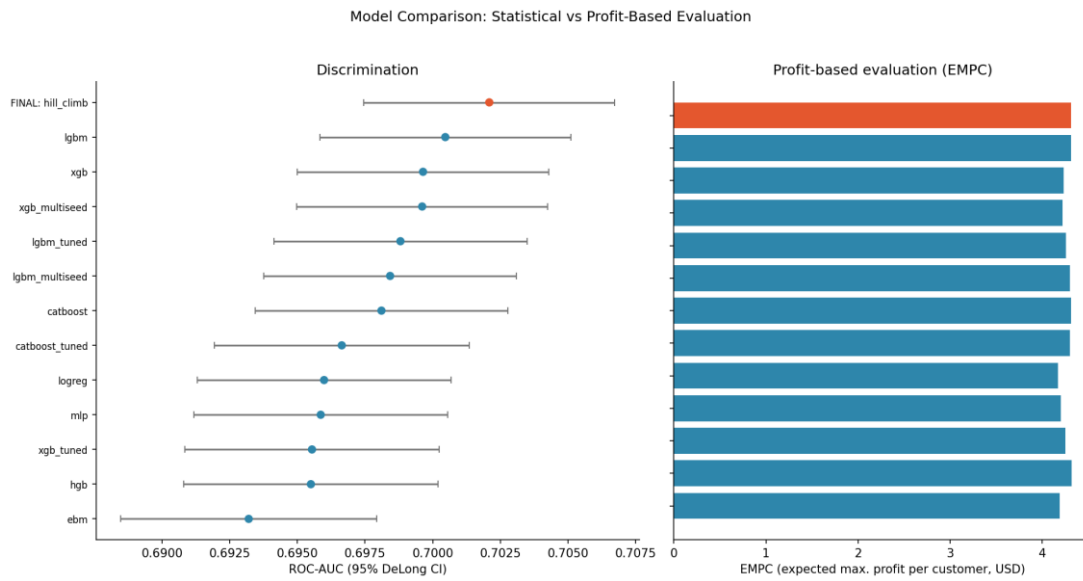


Figure F.3. Model Comparison: Statistical vs Profit-Based Evaluation

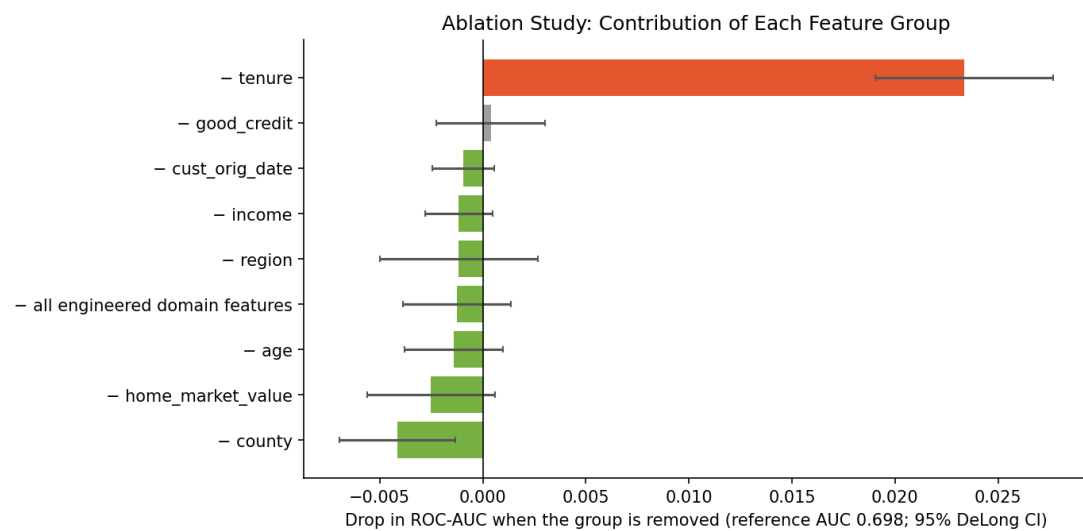


Figure F.4. Ablation Study: Contribution of Each Feature Group

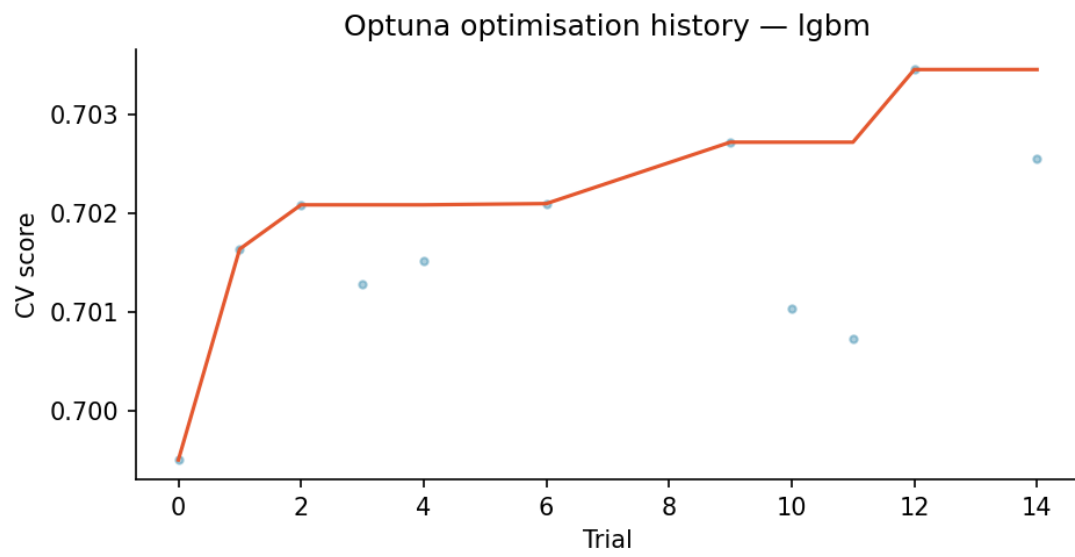


Figure F.5. Optuna optimisation history — lgbm

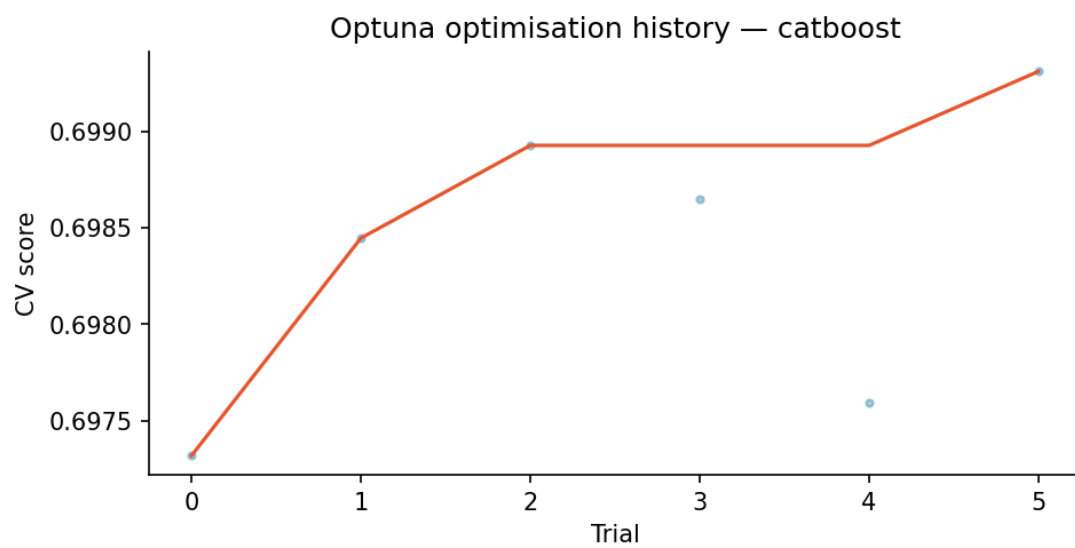


Figure F.6. Optuna optimisation history — catboost

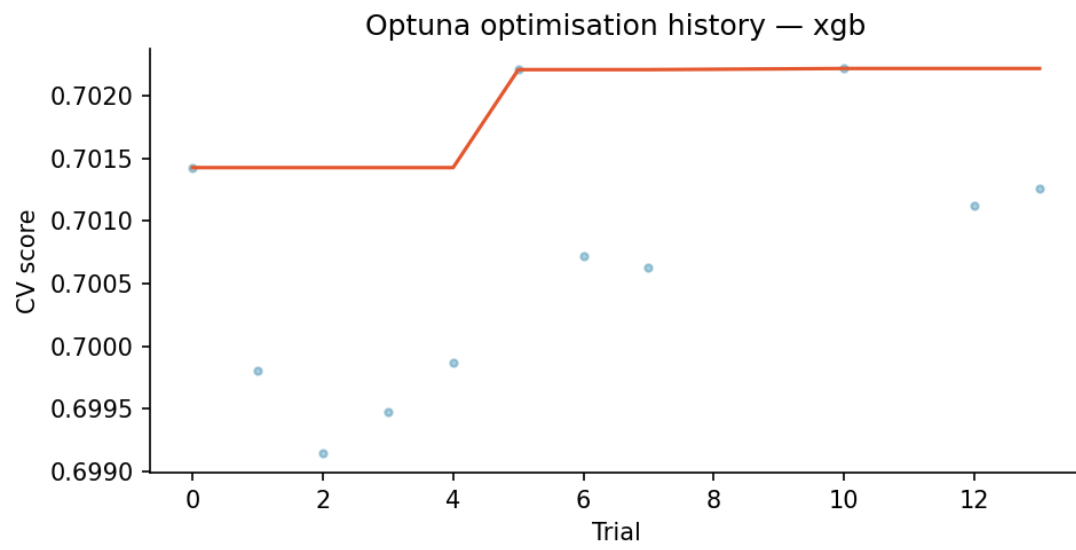


Figure F.7. Optuna optimisation history — xgb

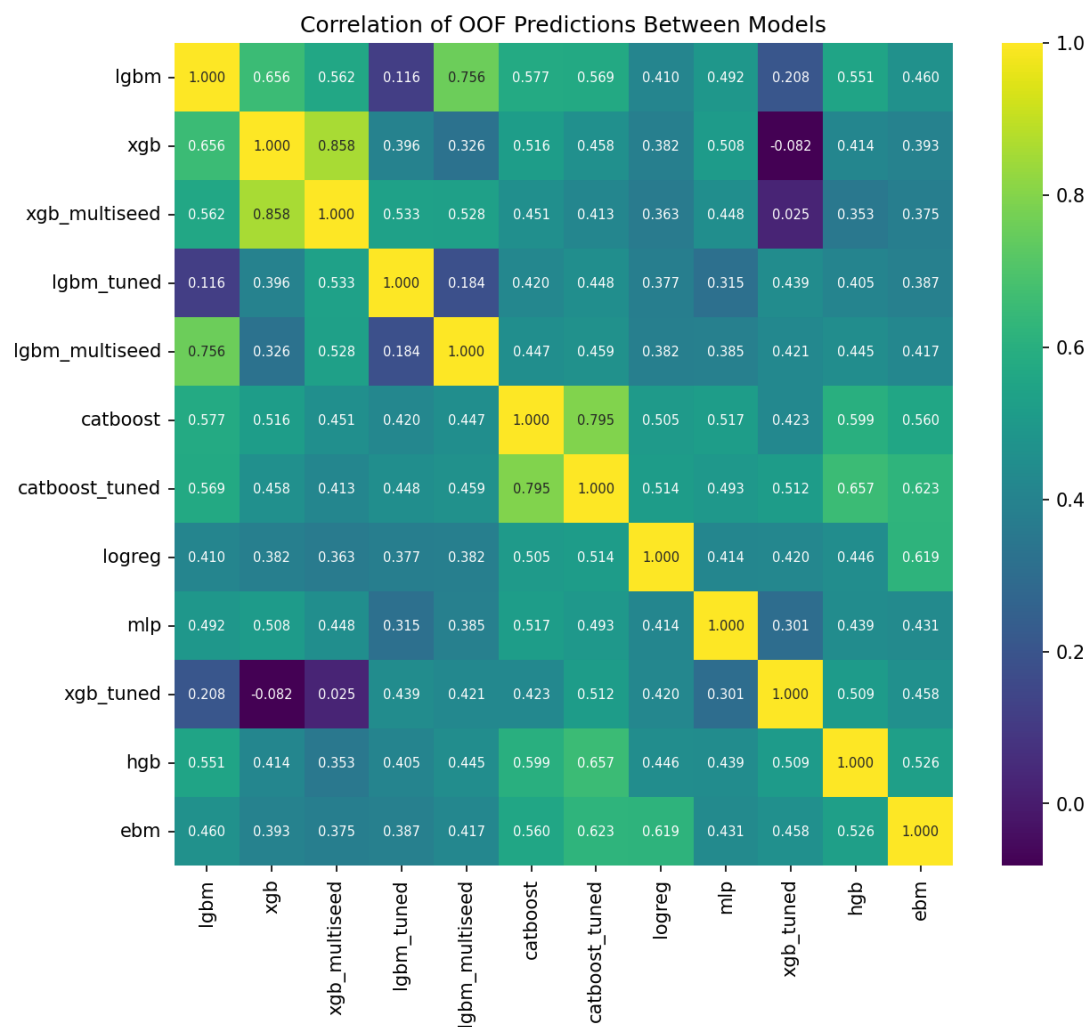


Figure F.8. Correlation of OOF Predictions Between Models

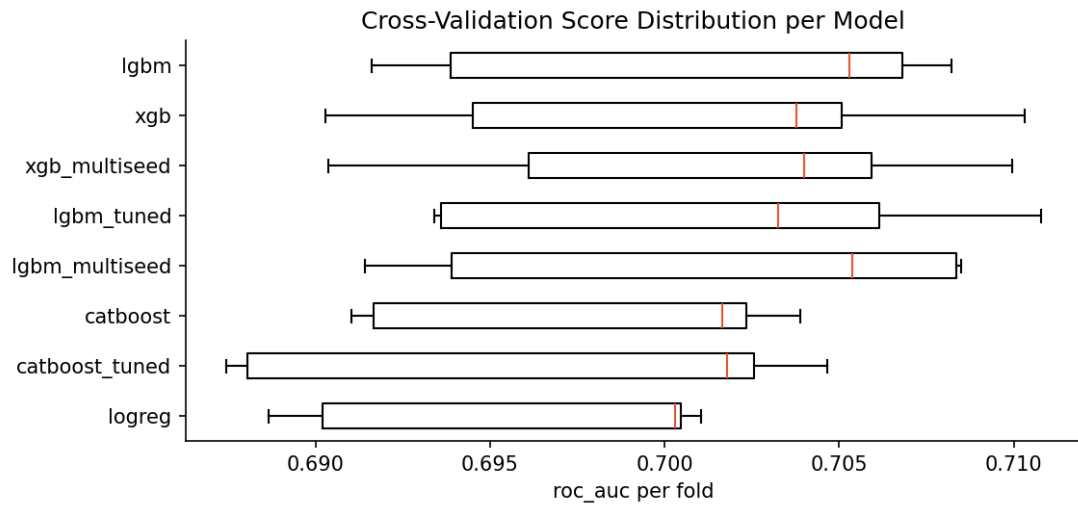


Figure F.9. Cross-Validation Score Distribution per Model

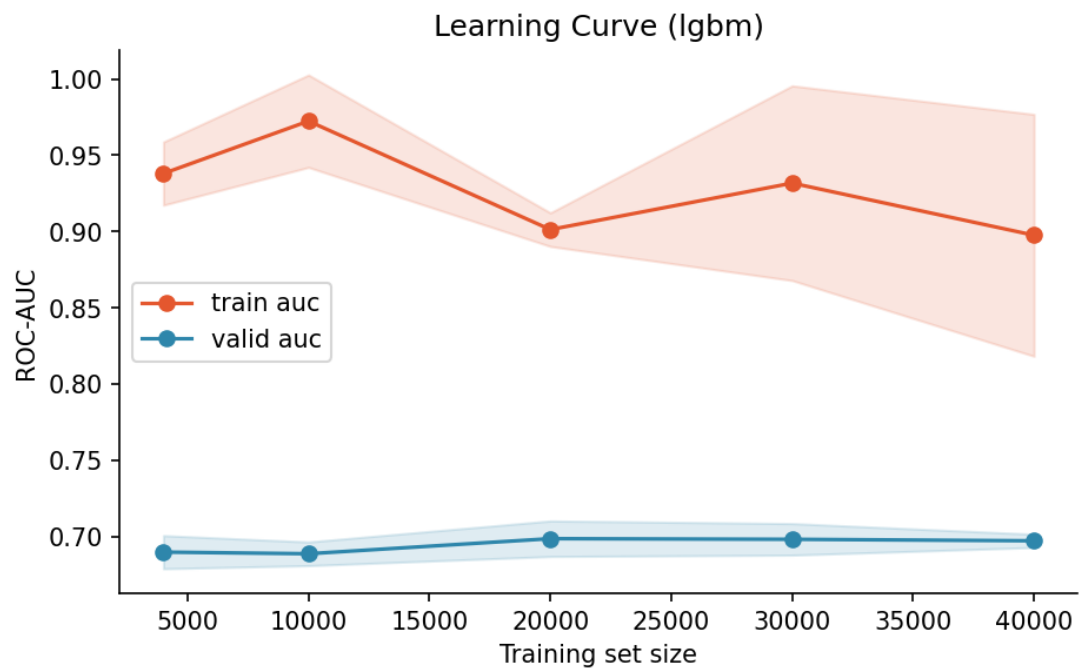


Figure F.10. Learning Curve (lgbm)

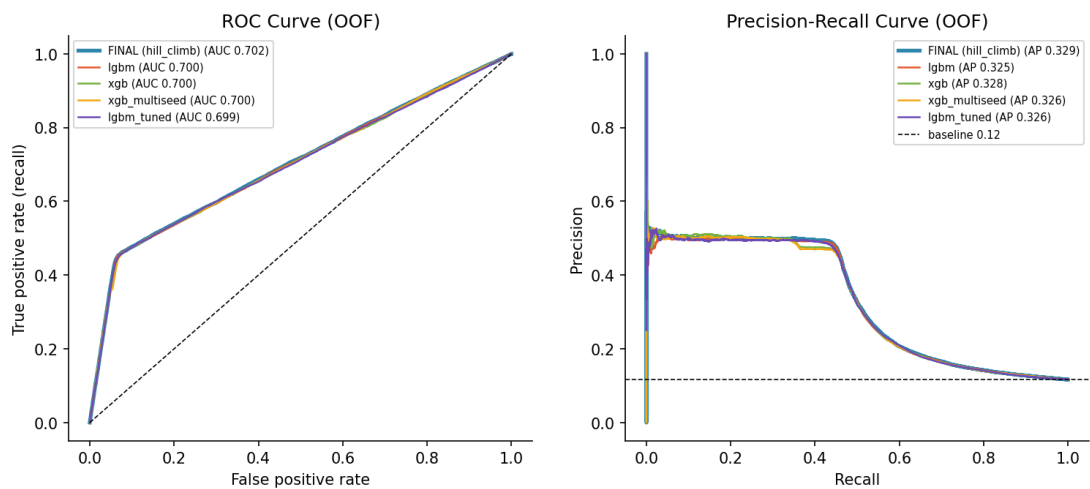


Figure F.11. ROC Curve (OOF)

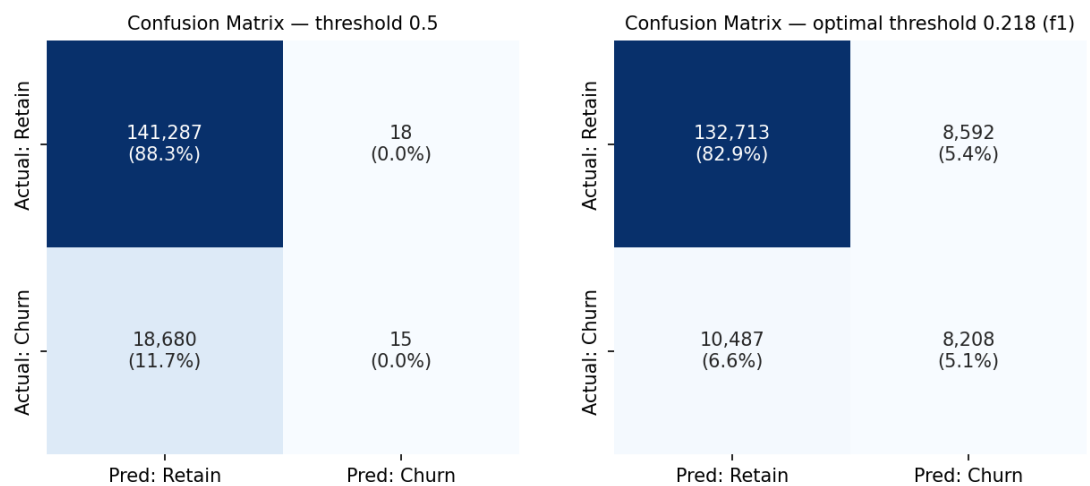


Figure F.12. Confusion Matrix — threshold 0.5

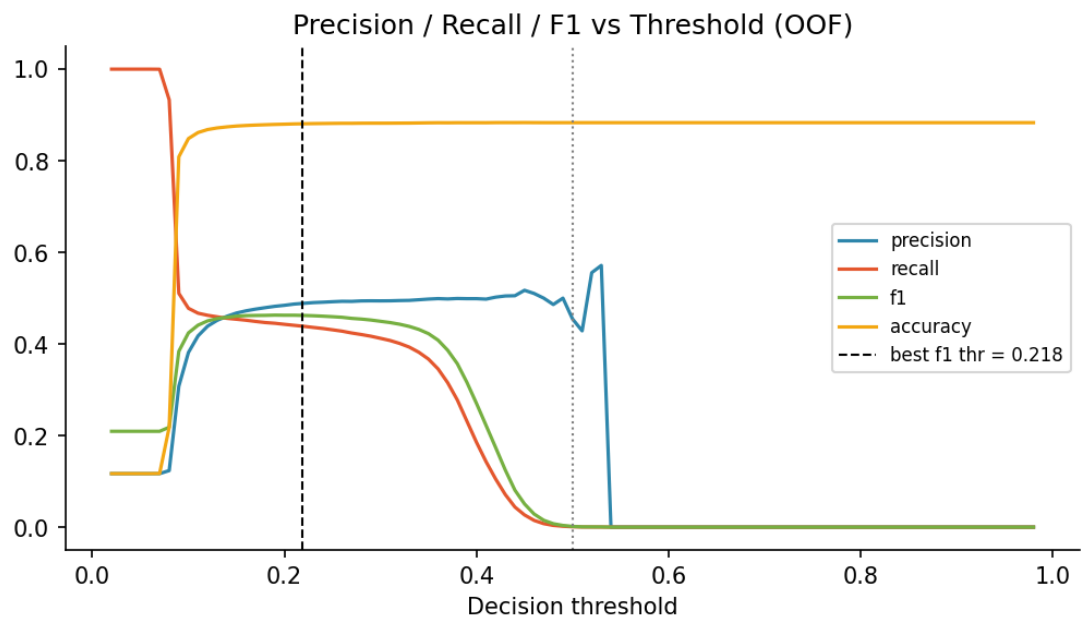


Figure F.13. Precision / Recall / F1 vs Threshold (OOF)

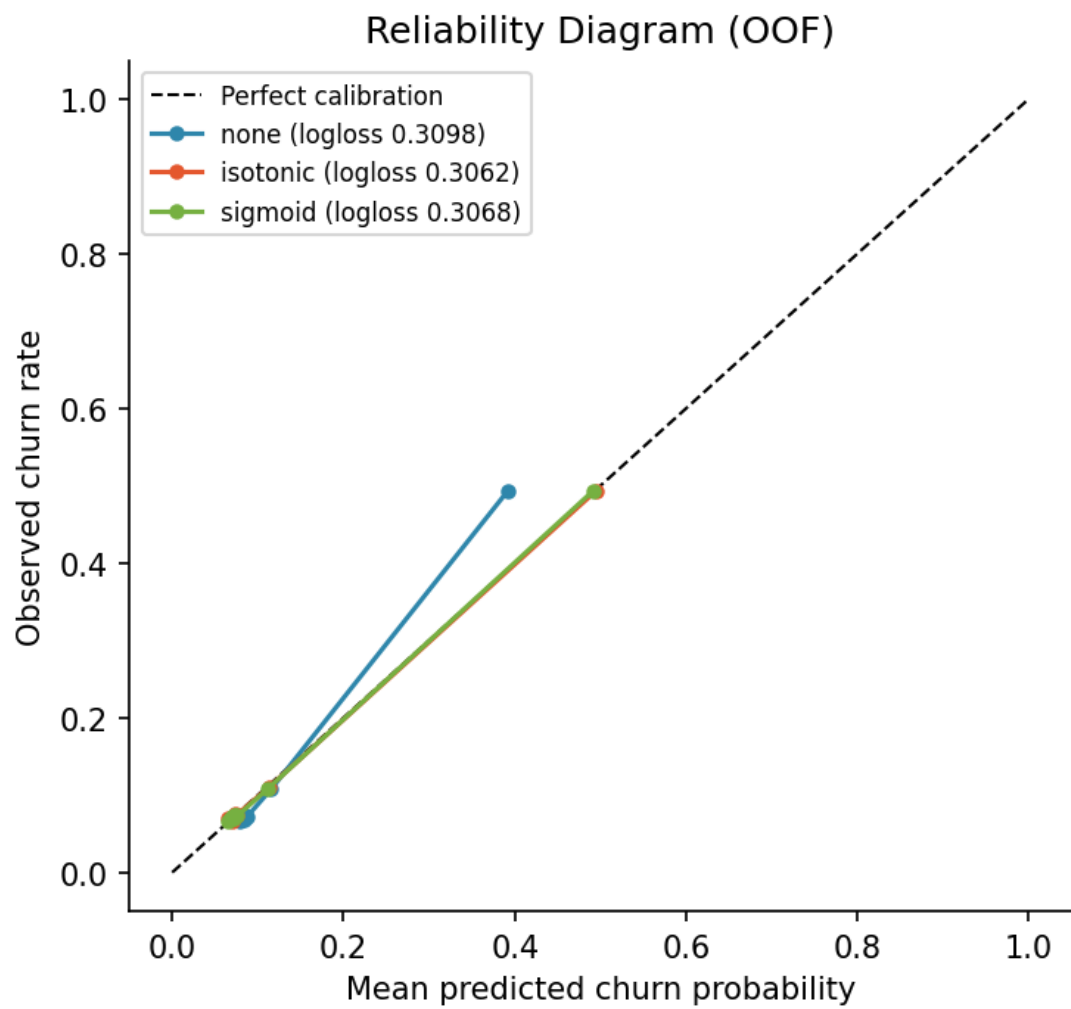


Figure F.14. Reliability Diagram (OOF)

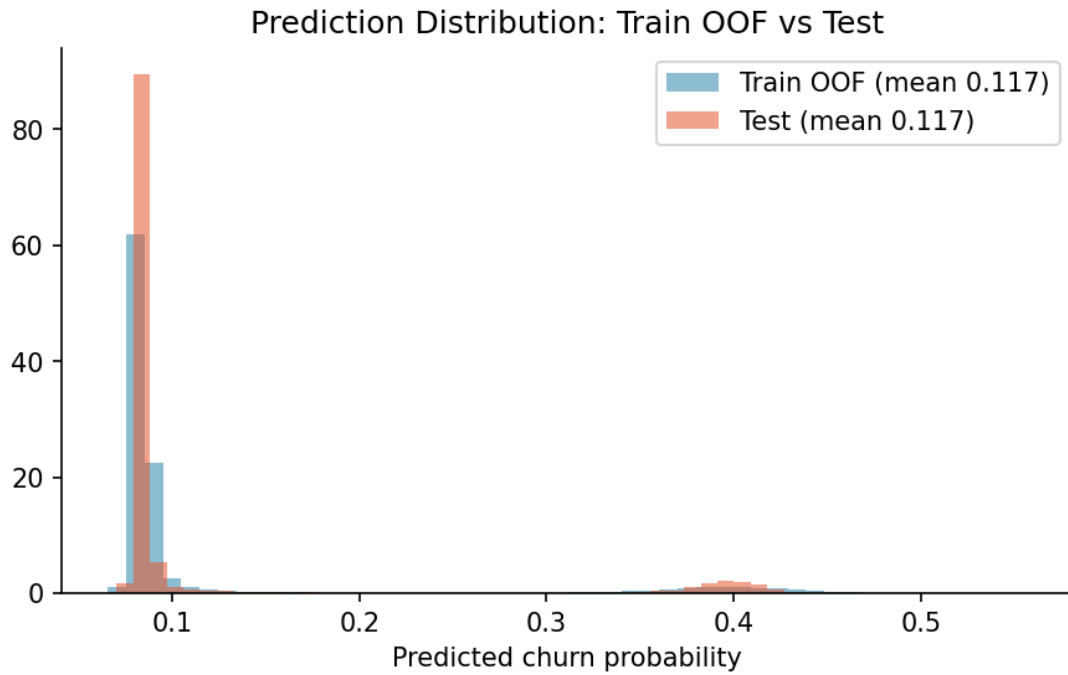


Figure F.15. Prediction Distribution: Train OOF vs Test

Table F.1. Highest-risk segments and the most loyal segment.

Segment (decision-tree rule)	Customers	Churn rate	Lift
tenure band = 0-6m AND cust orig date month ≥ 7 AND age in years ≤ 45	5,105	50.9%	4.4×
tenure band = 0-6m AND cust orig date month ≥ 7 AND age in years ≥ 46 AND curr ann amt ≤ 994.28	3,404	49.4%	4.2×
tenure band = 0-6m AND cust orig date month ≥ 7 AND age in years ≥ 46 AND curr ann amt > 994.28	3,367	48.6%	4.2×
tenure band $\neq$ 0-6m AND tenure band $\neq$ 6-12m AND tenure band $\neq$ 1-2y AND latitude ≤ 32.85	77,717	6.9%	0.6×

Table F.2. Full model comparison: AUC with DeLong CI, PR-AUC, Brier, ECE, KS and EMPC.

Model	Roc auc	Auc ci low	Auc ci high	Pr auc	Brier	Ece	Ks	Empe per customer	Delong p holm
FINAL: hill climb	0.702	0.697	0.707	0.329	0.088	0.021	0.386	4.312	
lgbm	0.700	0.696	0.705	0.325	0.088	0.013	0.386	4.309	0.210
xgb	0.700	0.695	0.704	0.328	0.090	0.022	0.383	4.231	0.149
xgb multiseed	0.700	0.695	0.704	0.326	0.092	0.032	0.383	4.220	0.135
lgbm tuned	0.699	0.694	0.703	0.326	0.090	0.027	0.383	4.253	0.141

Model	Roc auc	Auc ci low	Auc ci high	Pr auc	Brier	Ece	Ks	Empc per customer	Delong p holm
lgbm_multiseed	0.698	0.694	0.703	0.322	0.090	0.025	0.386	4.298	0.077
catboost	0.698	0.693	0.703	0.329	0.087	0.004	0.386	4.311	0.081
catboost tuned	0.697	0.692	0.701	0.327	0.087	0.004	0.386	4.299	0.012
logreg	0.696	0.691	0.701	0.326	0.088	0.008	0.375	4.171	0.013
mlp	0.696	0.691	0.701	0.323	0.088	0.013	0.379	4.200	0.008
xgb tuned	0.696	0.691	0.700	0.320	0.090	0.025	0.384	4.248	0.030
hgb	0.696	0.691	0.700	0.324	0.087	0.005	0.386	4.319	0.003
ebm	0.693	0.688	0.698	0.325	0.088	0.007	0.379	4.188	2.3e-05

Table F.3. Ablation study by feature group (DeLong tests, Holm-corrected) (first 10 columns; full table in report_tables.xlsx).

Ablation	N features removed	Features	Auc full	Auc without	Auc drop	Ci low	Ci high	Delong p	Holm p
– tenure	6	days tenure, tenure (years), new customer (<12 months), cust orig date (days since), cust orig date year, tenure band	0.698	0.674	0.023	0.019	0.028	1.7e-26	1.5e-25
– good credit	1	good credit	0.698	0.697	3.9e-04	-0.002	0.003	0.773	1.000
– cust orig date	1	cust orig date month	0.698	0.699	-9.6e-04	-0.002	5.4e-04	0.211	1.000
– income	1	income	0.698	0.699	-0.001	-0.003	4.7e-04	0.163	0.976
– region	1	city	0.698	0.699	-0.001	-0.005	0.003	0.550	1.000
– all engineered domain features	9	tenure (years), new customer (<12 months), premium-to-income ratio, cust orig date (days since), cust orig date year, cust orig date month, number of missing fields, tenure band	0.698	0.699	-0.001	-0.004	0.001	0.346	1.000
– age	2	age in years, age band	0.698	0.699	-0.001	-0.004	9.9e-04	0.251	1.000
– home market value	1	home market value	0.698	0.700	-0.003	-0.006	5.9e-04	0.113	0.791

Ablation	N features removed	Features	Auc full	Auc without	Auc drop	Ci low	Ci high	Delong p	Holm p
– county	1	county	0.698	0.702	-0.004	-0.007	-0.001	0.004	0.031

Table F.4. Ensemble methods compared.

Method	Oof roc auc	Oof auc	Weights
best_single	0.700	0.700	{'lgbm': 1.0}
mean_top3	0.701	0.701	{'lgbm': 0.333, 'xgb': 0.333, 'xgb_multiseed': 0.333}
rank_mean_top3	0.701	0.701	{'lgbm': 0.333, 'xgb': 0.333, 'xgb_multiseed': 0.333}
hill_climb	0.702	0.702	{'lgbm': 0.5, 'lgbm_tuned': 0.5}
optimized_weights	0.699	0.699	{'lgbm': 0.22, 'lgbm_tuned': 0.061, 'catboost': 0.207, 'catboost_tuned': 0.501, 'mlp': 0.011}
stacking_logreg	0.700	0.700	{'lgbm': 0.51, 'xgb': 0.249, 'xgb_multiseed': -0.167, 'lgbm_tuned': 0.238, 'lgbm_multiseed': -0.151, 'catboost': 0.6, 'catboost_tuned': -0.095, 'logreg': -0.191, 'mlp': 0.008, 'xgb_tuned': 0.16}

Table F.5. Bootstrap 95% confidence intervals for the final model.

Metric	Estimate	Ci low	Ci high
roc_auc	0.702	0.698	0.707
pr_auc	0.329	0.322	0.337
logloss	0.310	0.307	0.313

Table F.6. Experiment log (all logged runs).

Exp id	Stage	Experiment	Note	Roc auc	Pr auc	Logloss	Time s
1	1 baseline model zoo	dummy	Baseline (prior), default hyperparameters	0.500	0.117	0.361	2.589
2	1 baseline model zoo	logreg	Logistic regression, default hyperparameters	0.696	0.326	0.310	17.1
3	1 baseline model zoo	hgb	Hist. gradient boosting, default hyperparameters	0.696	0.324	0.307	24.8
4	1 baseline model zoo	lgbm	LightGBM, default hyperparameters	0.700	0.325	0.310	23.9
5	1 baseline model zoo	xgb	XGBoost, default hyperparameters	0.700	0.328	0.315	15.0
6	1 baseline model zoo	catboost	CatBoost, default hyperparameters	0.698	0.329	0.307	50.5
7	1 baseline model zoo	ebm	Explainable boosting machine	0.693	0.325	0.309	226.2

Exp id	Stage	Experiment	Note	Roc auc	Pr auc	Logloss	Time s
			(EBM), default hyperparameters				
8	1 baseline model zoo	mlp	Neural network (MLP), default hyperparameters	0.696	0.323	0.311	559.4
9	2 hyperparameter tuning	lgbm_tuned	Optuna TPE, 11 trials, 3-fold inner CV	0.699	0.326	0.314	26.9
10	2 hyperparameter tuning	catboost_tuned	Optuna TPE, 6 trials, 3-fold inner CV	0.697	0.327	0.307	34.0
11	2 hyperparameter tuning	xgb_tuned	Optuna TPE, 11 trials, 3-fold inner CV	0.696	0.320	0.316	11.9
12	3 multi-seed averaging	lgbm_multiseed	average of 2 seeds (base: lgbm)	0.698	0.322	0.315	40.6
13	3 multi-seed averaging	xgb_multiseed	average of 2 seeds (base: xgb)	0.700	0.326	0.321	25.4
14	4 feature selection	lgbm_fs	drop 9 features with permutation importance ≤ 0	0.696	0.310	0.319	28.6
15	5 ablation study	ablation reference (all features)	reference for ablation (same folds & params)	0.698	0.321	0.316	13.9
16	5 ablation study	ablation – tenure	drop 6 feature(s): days tenure, tenure (years), new customer (<12 months), cust orig date (days since)...	0.674	0.255	0.334	18.9
17	5 ablation study	ablation – age	drop 2 feature(s): age in years, age band	0.699	0.322	0.315	10.5
18	5 ablation study	ablation – county	drop 1 feature(s): county	0.702	0.326	0.311	10.8
19	5 ablation study	ablation – income	drop 1 feature(s): income	0.699	0.322	0.311	10.9
20	5 ablation study	ablation – home market value	drop 1 feature(s): home market value	0.700	0.326	0.308	11.0
21	5 ablation study	ablation – cust orig date	drop 1 feature(s): cust orig date month	0.699	0.318	0.315	11.4
22	5 ablation study	ablation – region	drop 1 feature(s): city	0.699	0.325	0.315	9.099
23	5 ablation study	ablation – good credit	drop 1 feature(s): good credit	0.697	0.321	0.311	12.5
24	5 ablation study	ablation – all engineered domain features	drop 9 feature(s): tenure (years), new customer (<12 months), premium-to-income ratio, cust orig date (days since)...	0.699	0.322	0.313	14.2
25	6 ensembling	ensemble: best single	lgbm 1.00	0.700	0.325	0.310	
26	6 ensembling	ensemble:	lgbm 0.33, xgb	0.701	0.329	0.313	

Exp id	Stage	Experiment	Note	Roc auc	Pr auc	Logloss	Time s
		mean_top3	0.33, xgb_multiseed 0.33				
27	6 ensembling	ensemble: rank_mean_top3	lgbm 0.33, xgb 0.33, xgb_multiseed 0.33	0.701	0.329	0.798	
28	6 ensembling	ensemble: hill_climb	lgbm 0.50, lgbm_tuned 0.50	0.702	0.329	0.310	
29	6 ensembling	ensemble: optimized_weights	lgbm 0.22, lgbm_tuned 0.06, catboost 0.21, catboost_tuned 0.50, mlp 0.01	0.699	0.329	0.307	
30	6 ensembling	ensemble: stacking_logreg	lgbm 0.51, xgb 0.25, xgb_multiseed - 0.17, lgbm_tuned 0.24, lgbm_multiseed -0.15, catboost 0.60, catboost_tuned - 0.10, logreg - 0.19, mlp 0.01, xgb_tuned 0.16	0.700	0.328	0.306	

Table F.7. Train–test drift (PSI, KS; top 15).

Feature	Psi	Ks stat	Ks p	N unseen levels	Unseen rows %	Psi flag
city	0.003			0	0	stable
length of residence	4.9e-04	0.005	0.409			stable
latitude	4.3e-04	0.007	0.193			stable
longitude	3.2e-04	0.006	0.249			stable
curr ann amt	2.8e-04	0.004	0.686			stable
days tenure	2.4e-04	0.004	0.545			stable
cust orig date (days since)	2.4e-04	0.004	0.545			stable
tenure (years)	2.4e-04	0.004	0.545			stable
county	2.3e-04			0	0	stable
premium-to- income ratio	2.1e-04	0.004	0.788			stable
age in years	2.1e-04	0.005	0.399			stable
cust orig date month	1.7e-04	0.002	0.997			stable
cust orig date year	1.5e-04	0.004	0.752			stable
home owner	1.5e-04			0	0	stable
income	1.4e-04	0.003	0.973			stable

Appendix G. Explainability

Global and local explanations: gain, permutation and SHAP importance, dependence and partial dependence, individual explanations and segment heterogeneity.

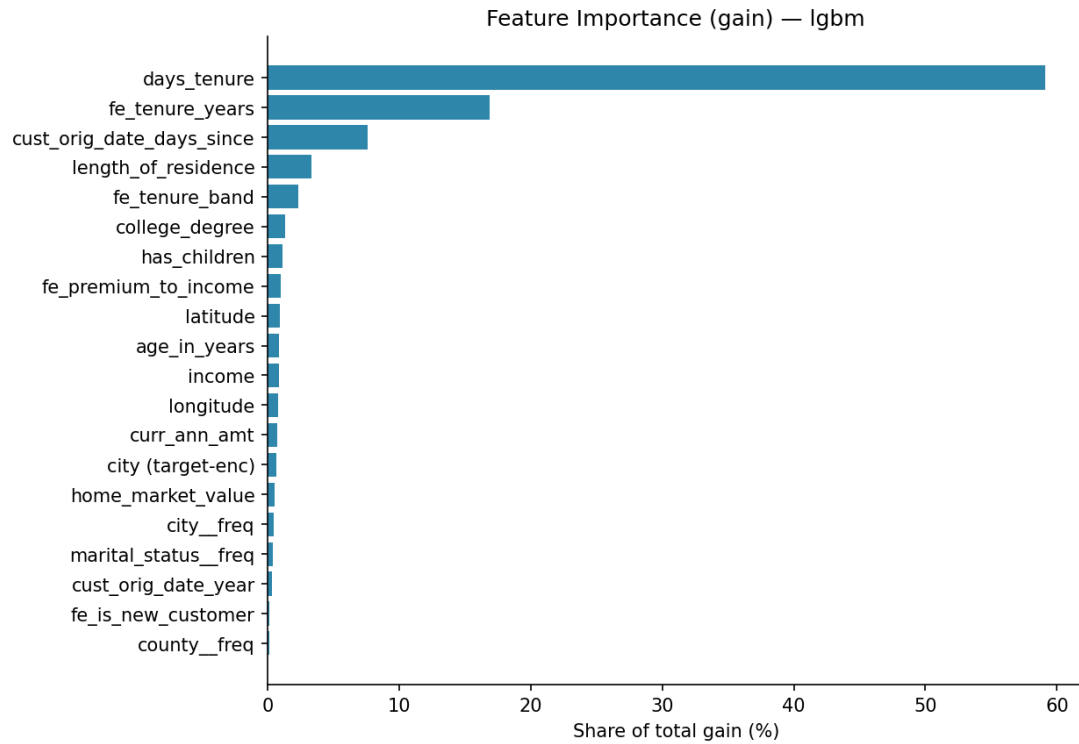


Figure G.1. Feature Importance (gain) — lgbm

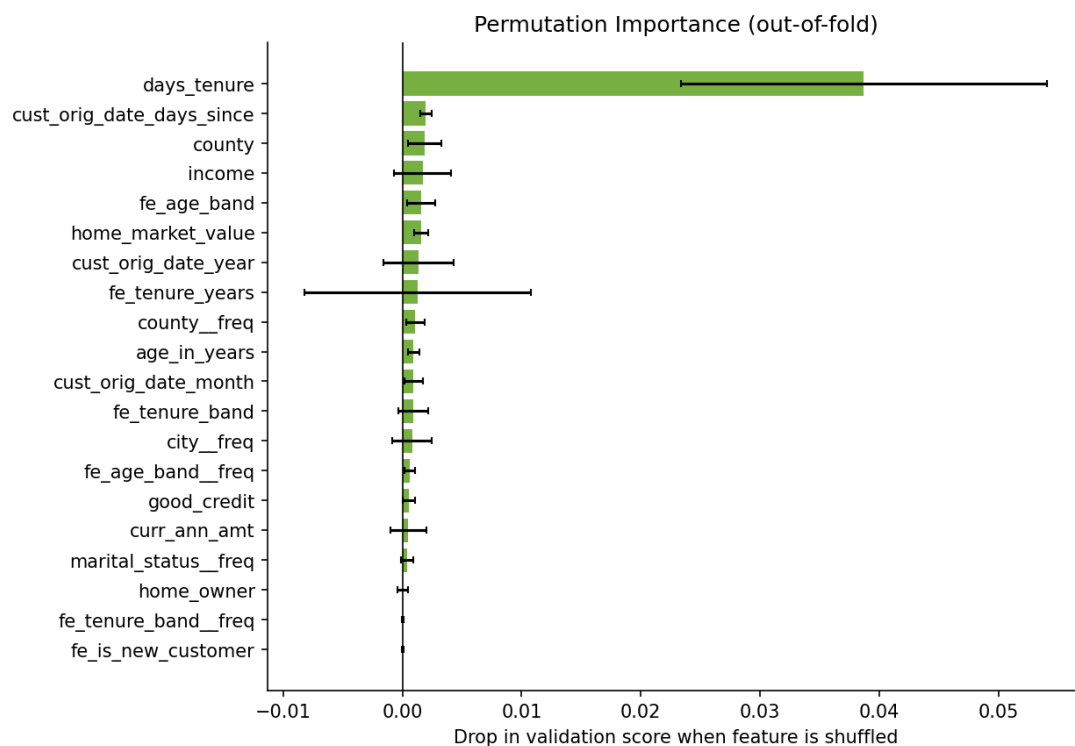


Figure G.2. Permutation Importance (out-of-fold)

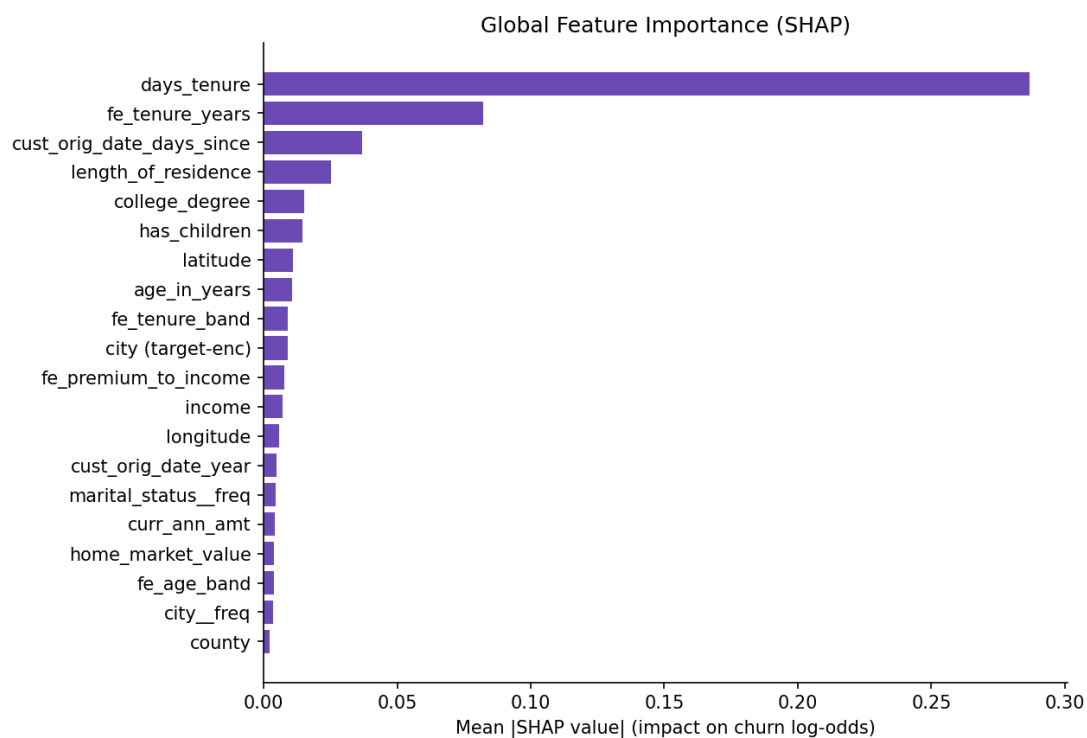


Figure G.3. Global Feature Importance (SHAP)

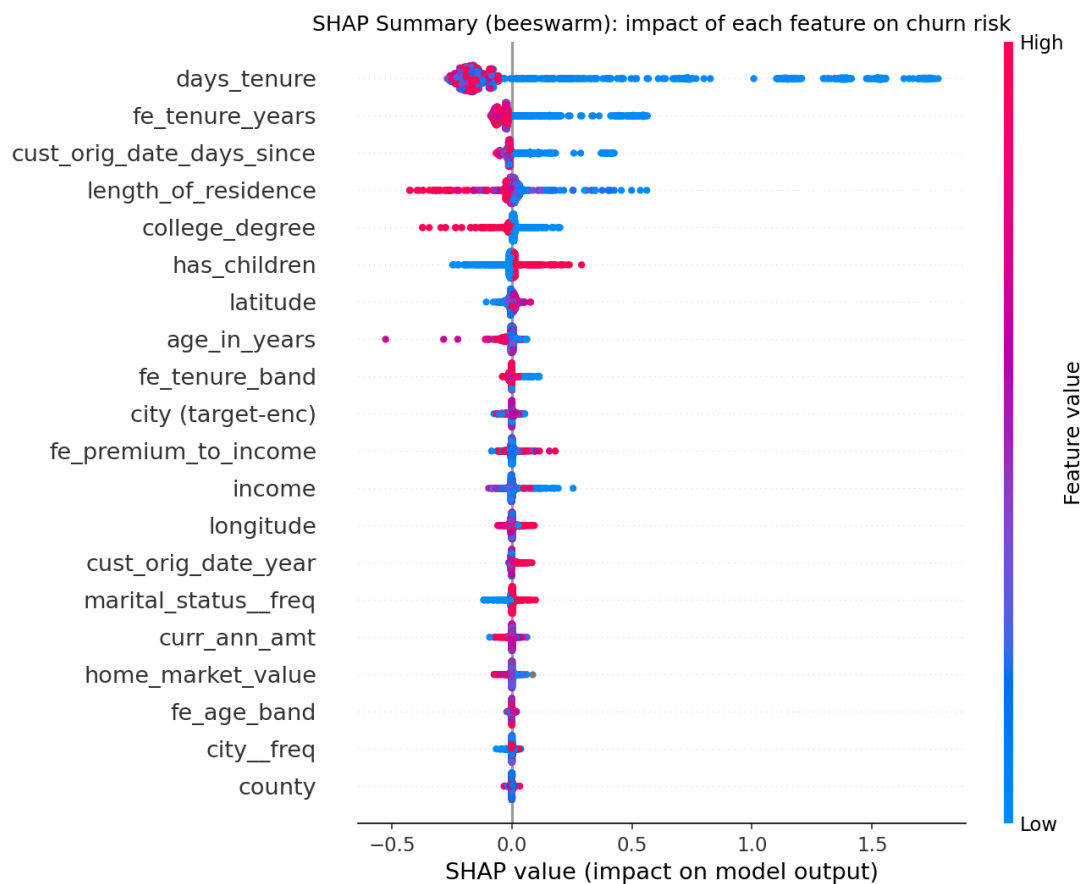


Figure G.4. SHAP summary: each dot is a customer; positive values push churn risk up (colour = feature value).

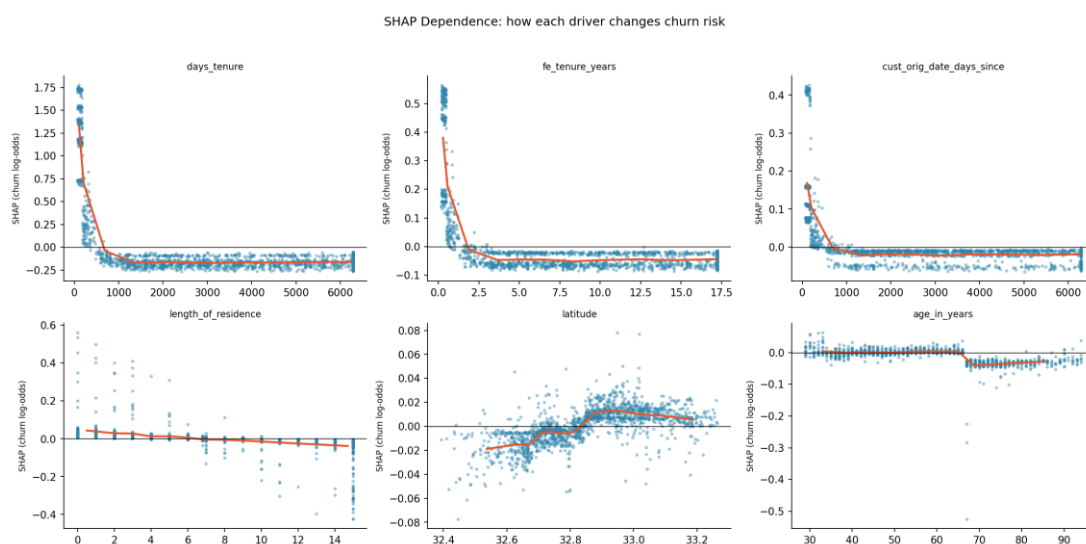


Figure G.5. SHAP Dependence: how each driver changes churn risk

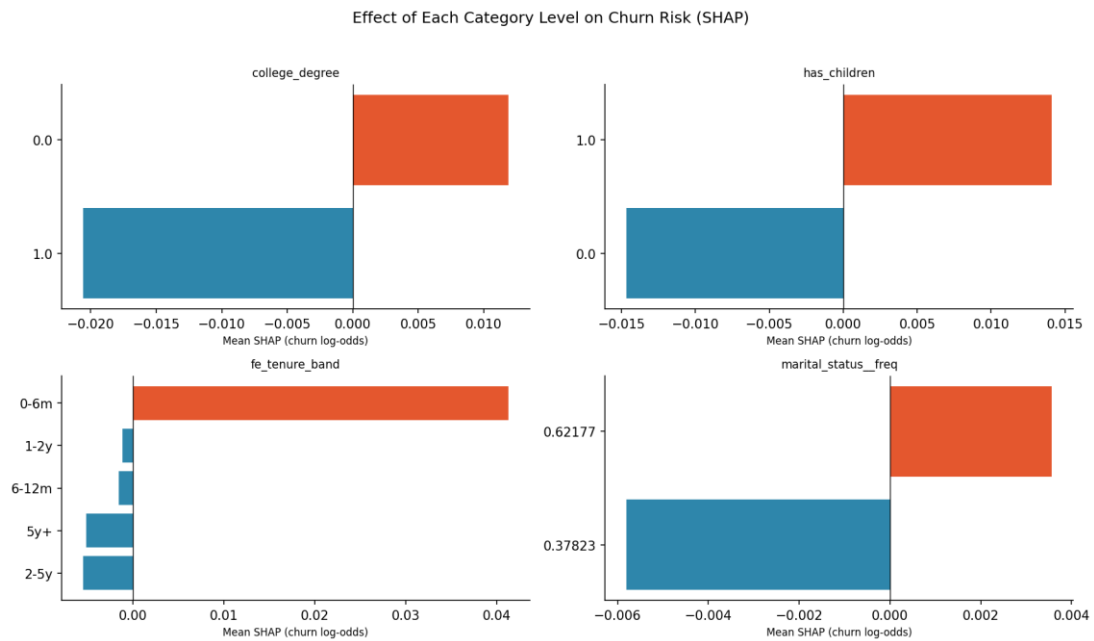


Figure G.6. Effect of Each Category Level on Churn Risk (SHAP)

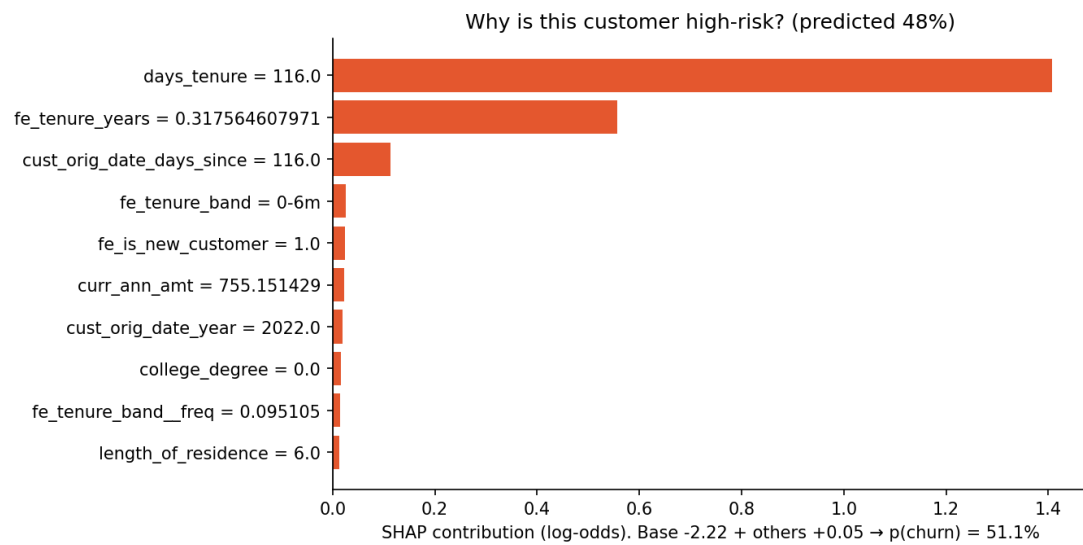


Figure G.7. Why is this customer high-risk? (predicted 48%)

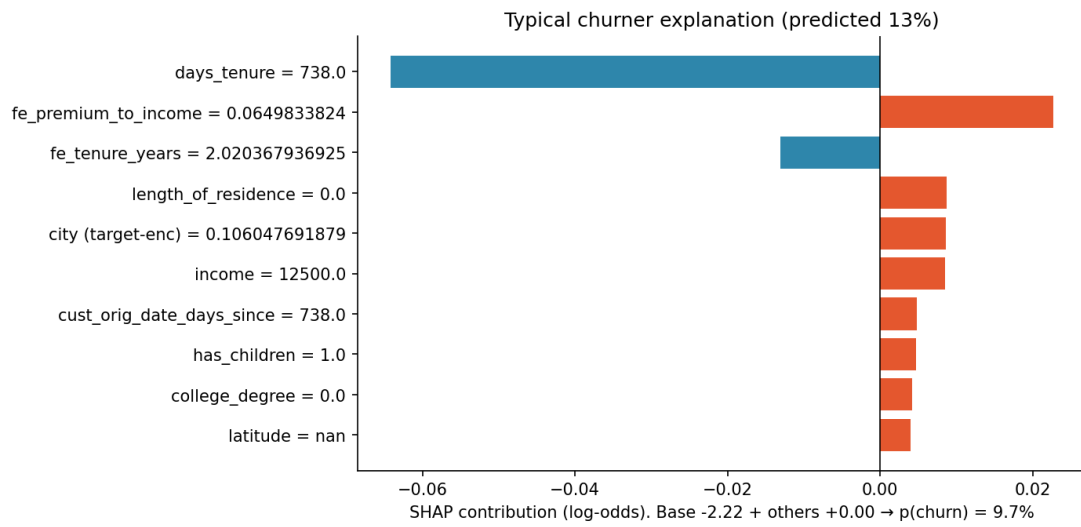


Figure G.8. Typical churner explanation (predicted 13%)

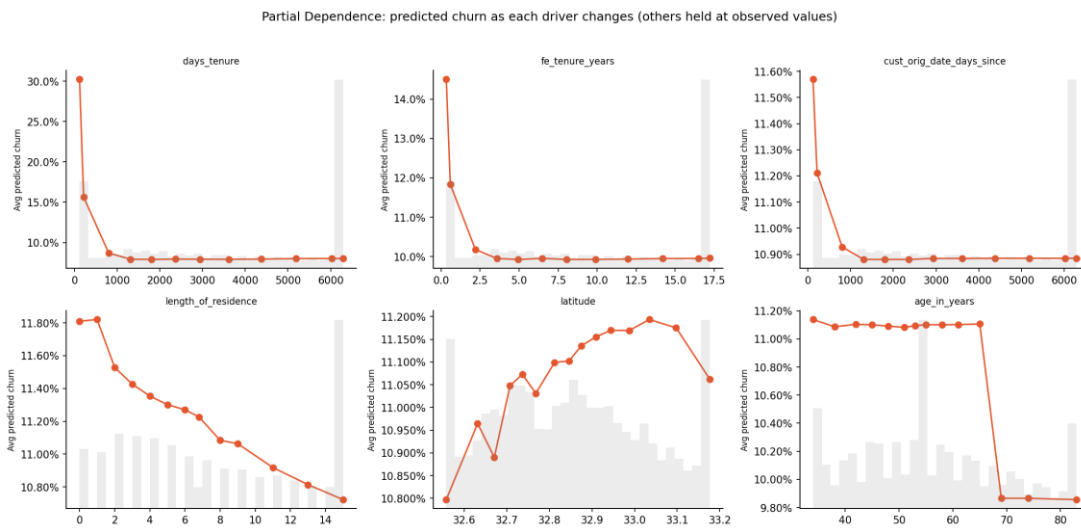


Figure G.9. Partial Dependence: predicted churn as each driver changes (others held at observed values)

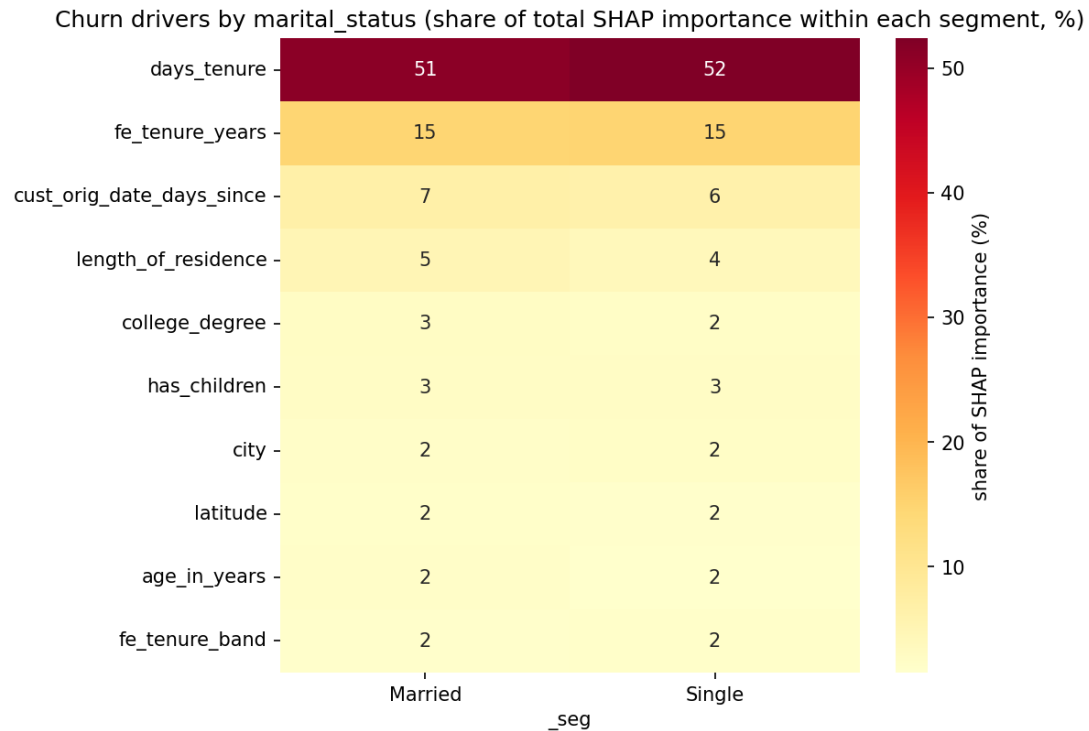


Figure G.10. Churn drivers by marital status (share of total SHAP importance within each segment, %)

Table G.1. SHAP importance and direction of effect (top 20).

Feature	Mean abs shap	Share %	Direction
days tenure	0.287	51.6	higher value → lower churn risk
tenure (years)	0.082	14.8	higher value → lower churn risk
cust orig date (days since)	0.037	6.609	higher value → lower churn risk
length of residence	0.025	4.538	higher value → lower churn risk
college degree	0.015	2.696	riskiest: '0.0' (+0.01); safest: '1.0' (-0.02)
has children	0.014	2.587	riskiest: '1.0' (+0.01); safest: '0.0' (-0.01)
latitude	0.011	1.973	higher value → higher churn risk
age in years	0.011	1.929	higher value → lower churn risk
tenure band	0.009	1.626	riskiest: '0-6m' (+0.04); safest: '2-5y' (-0.01)
city (target-enc)	0.009	1.607	higher value → higher churn risk
premium-to-income ratio	0.008	1.360	non-monotonic / mixed
income	0.007	1.293	non-monotonic / mixed
longitude	0.006	1.019	higher value → higher churn risk
cust orig date year	0.005	0.864	higher value → higher churn risk
marital status_freq	0.004	0.804	riskiest: '0.62177' (+0.00);

Feature	Mean abs shap	Share %	Direction
			safest: '0.37823' (-0.01)
curr ann amt	0.004	0.744	higher value → lower churn risk
home market value	0.004	0.713	higher value → lower churn risk
age band	0.004	0.695	riskiest: '55-64' (+0.00); safest: '65+' (-0.00)
city__freq	0.003	0.607	higher value → higher churn risk
county	0.002	0.415	riskiest: 'Dallas' (+0.00); safest: 'Parker' (-0.00)

Table G.2. Consolidated driver evidence matrix (first 10 columns; full table in report_tables.xlsx).

Feature	Iv	Effect size	Q value BH	Highest risk level	Highest risk rate	Odds ratio	Hazard ratio	Shap rank	Evidence score
tenure band	0.885	0.381	0	0-6m	0.488	12.2		10	3
age band	0.038	0.060	3.1e-120	25-34	0.146	0.921	0.484	16	3
days tenure	0.855	-0.391	0	20-191	0.473			1	2
tenure (years)	0.855	-0.391	0	0.0548-0.523	0.473			2	2
cust orig date (days since)	0.855	-0.391	0	20-191	0.473			3	2
length of residence	0.011	-0.061	1.9e-41	0-1	0.135			4	2
college degree	0.004	0.021	5.4e-17	0	0.122			5	2
has children	0.005	0.022	1.2e-17	1	0.123			6	2
age in years	0.038	-0.094	3.8e-96	23-37	0.143			9	2
latitude	0.002	0.009	0.082	Missing	0.124			8	1
premium-to-income ratio	0.002	0.017	1.7e-04	0.0182-0.0232	0.124			11	1
cust orig date year	0.713	0.378	0	2,020-2,022	0.348			15	1
curr ann amt	0.003	0.030	3.8e-11	1,261-2,108	0.126			17	1
new customer (<12 months)	0.759	0.338	0	1	0.384			20	1
cust orig date month	0.086	-0.039	1.2e-18	6-7	0.201	1.011		21	1
number of missing fields	0.002	0.011	1.0e-04	≥3	0.136			23	1
home owner	0.001	0.012	3.4e-06	0	0.125			24	1
income	4.3e-04	-0.003	0.624	27.5K-37.5K	0.120			12	0
marital status	1.7e-04	0.004	0.113	Single	0.119			13	0
longitude	0.002	-0.001	0.772	Missing	0.124			14	0
home market value	0.002	-0.004	0.433	Missing	0.134			18	0
county	6.3e-04	0.008	0.724	Rockwall	0.124			19	0

Feature	Iv	Effect size	Q value BH	Highest risk level	Highest risk rate	Odds ratio	Hazard ratio	Shap rank	Evidence score
good credit	2.6e-04	0.005	0.052	0	0.121			22	0

Appendix I. Business Analytics

Lift and gains, risk tiers, risk $\times$ value prioritisation, campaign simulation and sensitivity, churn personas.

Customer Churn — Executive Summary

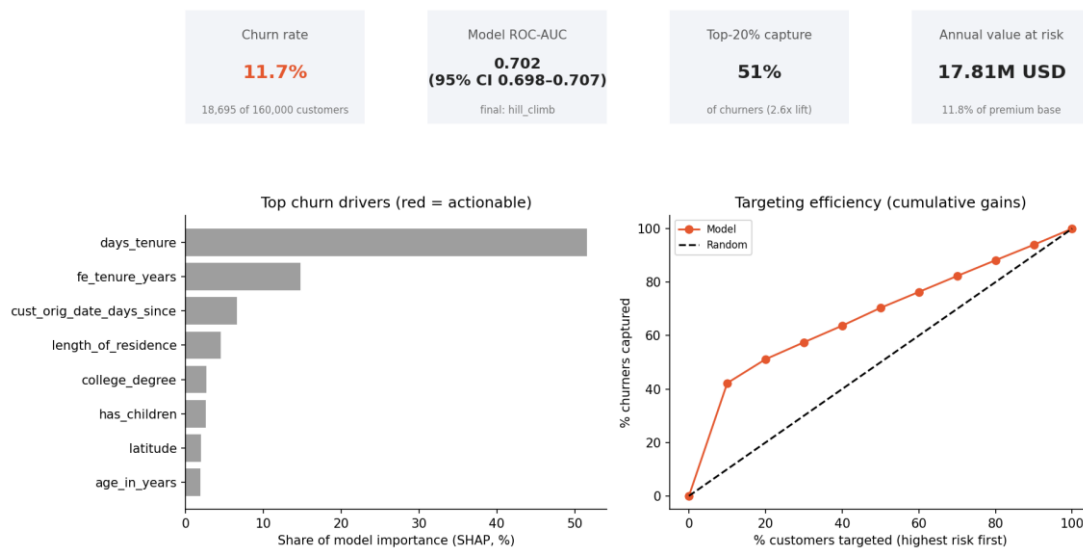


Figure I.1. Customer Churn — Executive Summary

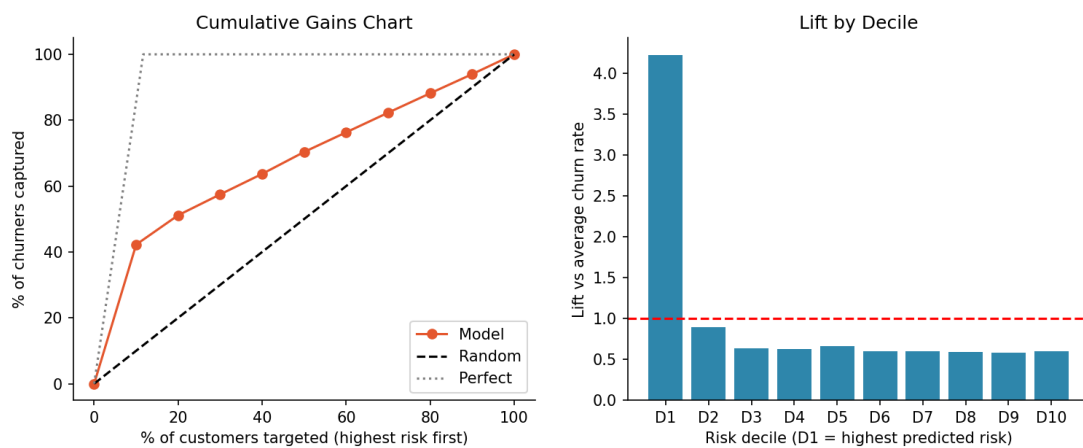


Figure I.2. Cumulative Gains Chart

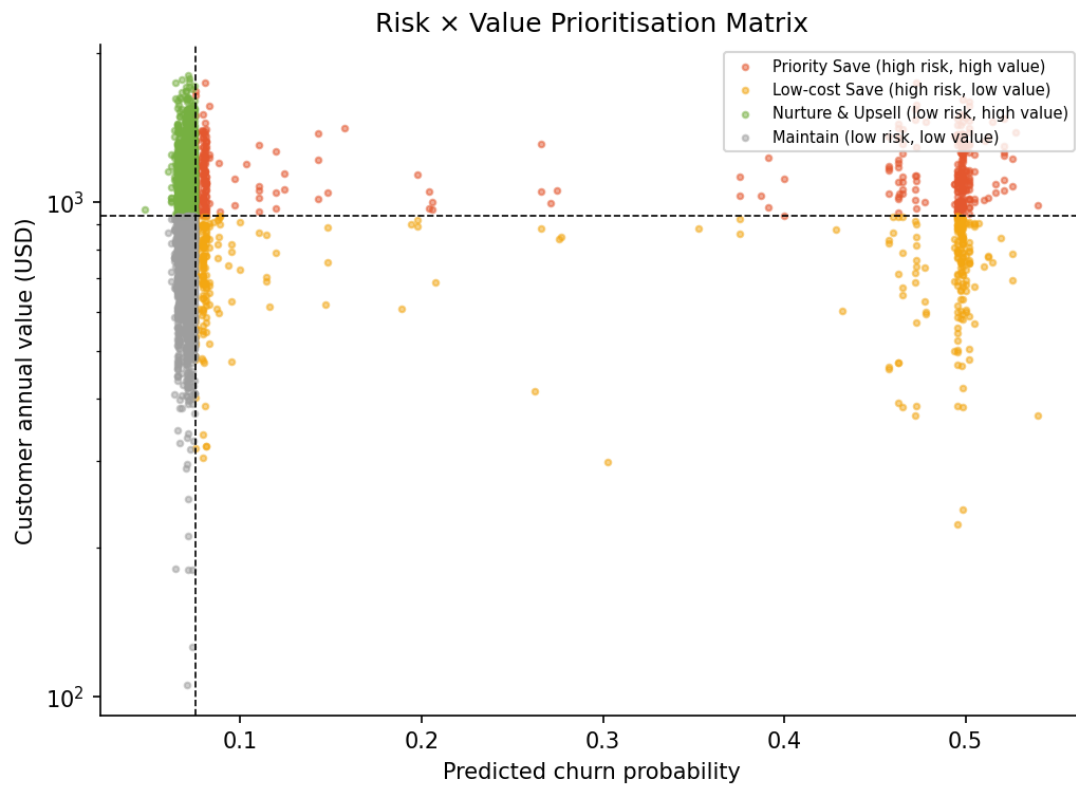


Figure I.3. Risk x Value Prioritisation Matrix

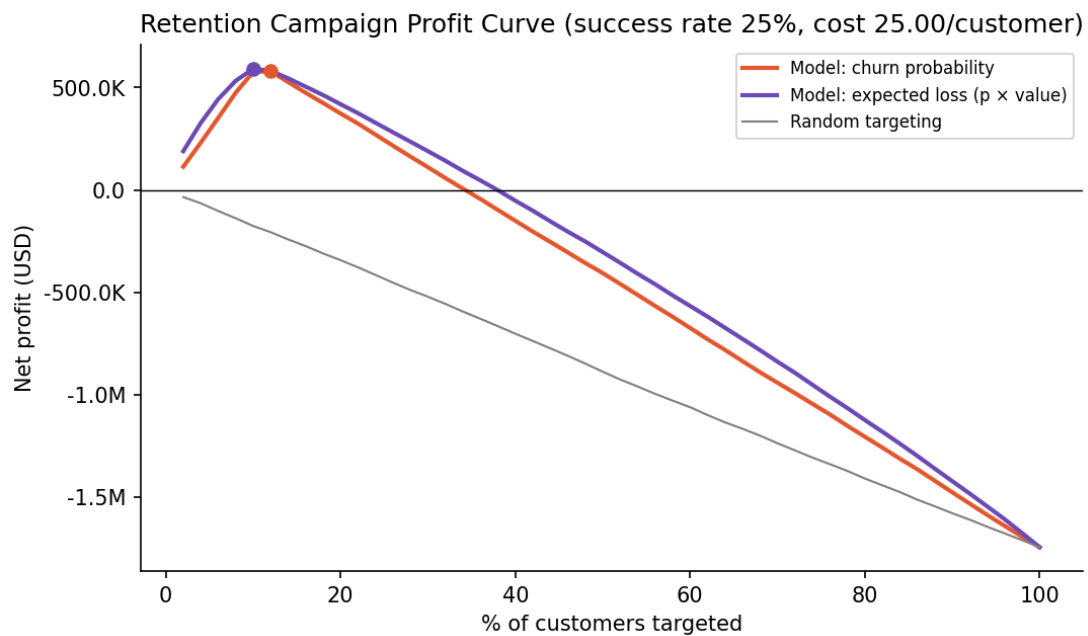


Figure I.4. Retention Campaign Profit Curve (success rate 25%, cost 25.00/customer)

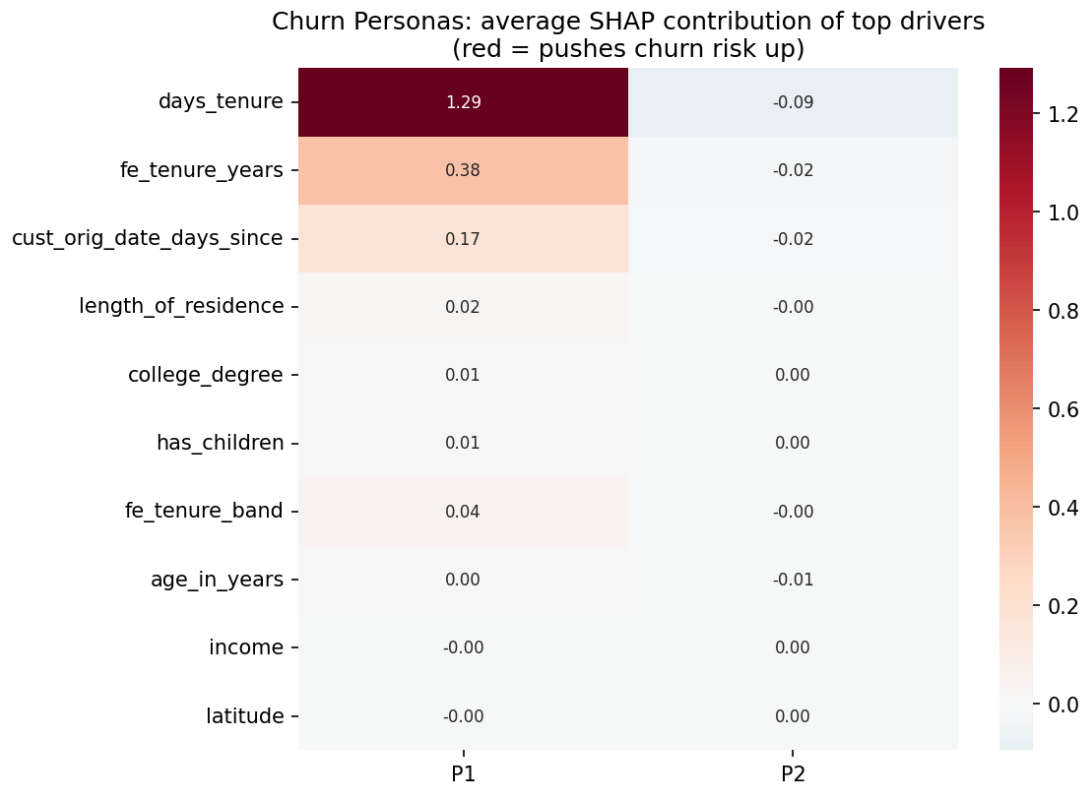


Figure I.5. Churn Personas: average SHAP contribution of top drivers
(red = pushes churn risk up)

Table I.1. Lift by risk decile.

Decile	N	Churners	Avg pred	Min pred	Churn rate	Lift	Cum customers %	Cum churners captured %	Cum lift
D1	16,000	7,896	0.494	0.458	0.493	4.224	10.0	42.2	4.224
D2	16,000	1,666	0.108	0.075	0.104	0.891	20.0	51.1	2.557
D3	16,000	1,183	0.075	0.074	0.074	0.633	30.0	57.5	1.916
D4	16,000	1,164	0.074	0.074	0.073	0.623	40.0	63.7	1.593
D5	16,000	1,245	0.073	0.072	0.078	0.666	50.0	70.4	1.407
D6	16,000	1,114	0.071	0.071	0.070	0.596	60.0	76.3	1.272
D7	16,000	1,116	0.071	0.071	0.070	0.597	70.0	82.3	1.176
D8	16,000	1,100	0.070	0.070	0.069	0.588	80.0	88.2	1.102
D9	16,000	1,085	0.068	0.066	0.068	0.580	90.0	94.0	1.044
D10	16,000	1,126	0.065	1.0e-04	0.070	0.602	100.0	100.0	1.000

Table I.2. Risk tiers.

Tier	N train	Actual churn rate	Avg predicted	Share churners %	N test
High	35,105	0.280	0.281	52.6	6,581
Medium	45,048	0.074	0.074	17.9	12,982
Low	79,847	0.069	0.069	29.5	20,437

Table I.3. Risk × value quadrants.

Quadrant	N	Actual churn rate	Avg pred	Total value	Value at risk	Share value at risk %
Low-cost Save (high risk, low value)	16,870	0.266	0.266	12.59M	3.36M	18.9
Maintain (low risk, low value)	63,128	0.072	0.071	46.94M	3.32M	18.7
Nurture & Upsell (low risk, high value)	61,767	0.070	0.071	70.21M	4.97M	27.9
Priority Save (high risk, high value)	18,235	0.293	0.295	20.78M	6.16M	34.6

Table I.4. Optimal campaign size by strategy.

Strategy	Target %	N targeted	Churners reached	Benefit	Cost	Net profit	Roi
Model: churn probability	12.0	19,200	8,597	1.06M	480,000	578,864	1.206
Model: expected loss (p × value)	10.0	16,000	7,809	990,809	400,000	590,809	1.477
Random targeting	2.000	3,200	374.0	45,029	80,000	-34,971	-0.437

Table I.5. Campaign sensitivity analysis.

Success rate	Cost per contact	Optimal target %	Max net profit	Roi
0.125	12.5	10.2	296,717	1.459
0.125	25.0	6.083	102,486	0.421
0.125	50.0	2.000	-25,579	-0.160
0.250	12.5	14.2	823,789	2.891
0.250	25.0	10.2	593,435	1.459
0.250	50.0	6.083	204,971	0.421
0.375	12.5	63.3	1.55M	1.227
0.375	25.0	10.2	1.09M	2.689
0.375	50.0	10.2	686,827	0.844

Table I.6. Churn personas (first 10 columns; full table in report_tables.xlsx).

Persona	N in sample	Share of at risk %	Avg pred churn	Actual churn rate	Top drivers	Tenure band	Age band	Days tenure	Tenure (years)
P2	791	78.6	0.083	0.085	no dominant driver (moderate, diffuse risk)	5y+	45-54	3,674	10.1
P1	216	21.4	0.490	0.519	days tenure (+1.29); tenure (years) (+0.38); cust orig date (days since) (+0.17)	0-6m	45-54	120.4	0.330

Appendix J. Governance, Deployment and Implementation

Fairness audit, model card, deployment loop, pilot design and implementation roadmap.

The model card follows Mitchell et al. (2019). The fairness audit compares calibration (mean predicted vs actual churn), discrimination (AUC) and selection into the High-risk tier across customer groups; differences in selection that mirror genuine differences in churn are expected, whereas calibration gaps would indicate bias. The score is restricted to retention outreach.

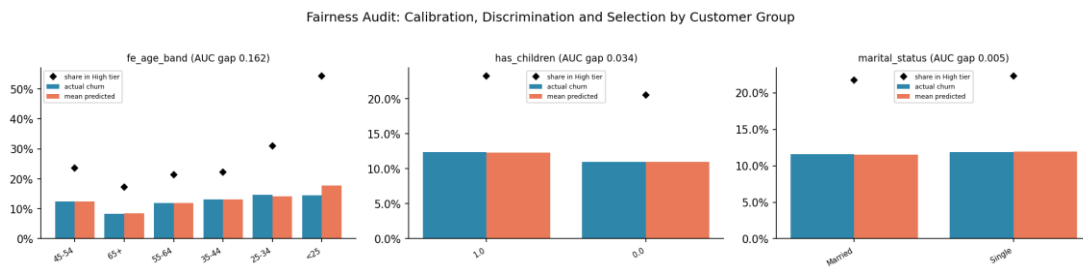


Figure J.1. Fairness Audit: Calibration, Discrimination and Selection by Customer Group

Deployment & Continuous-Learning Loop (MLOps)

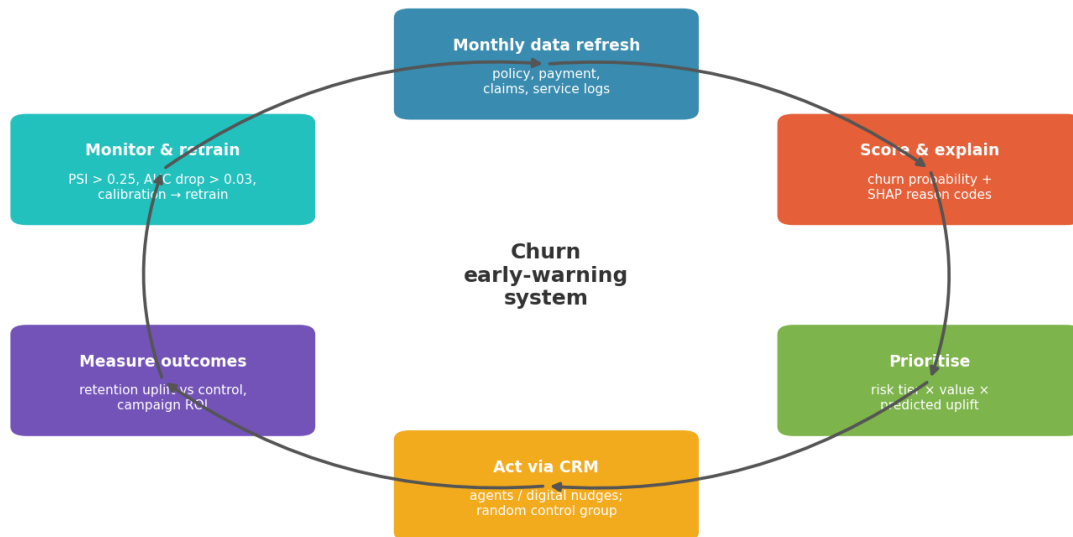


Figure J.2. Deployment & Continuous-Learning Loop (MLOps)

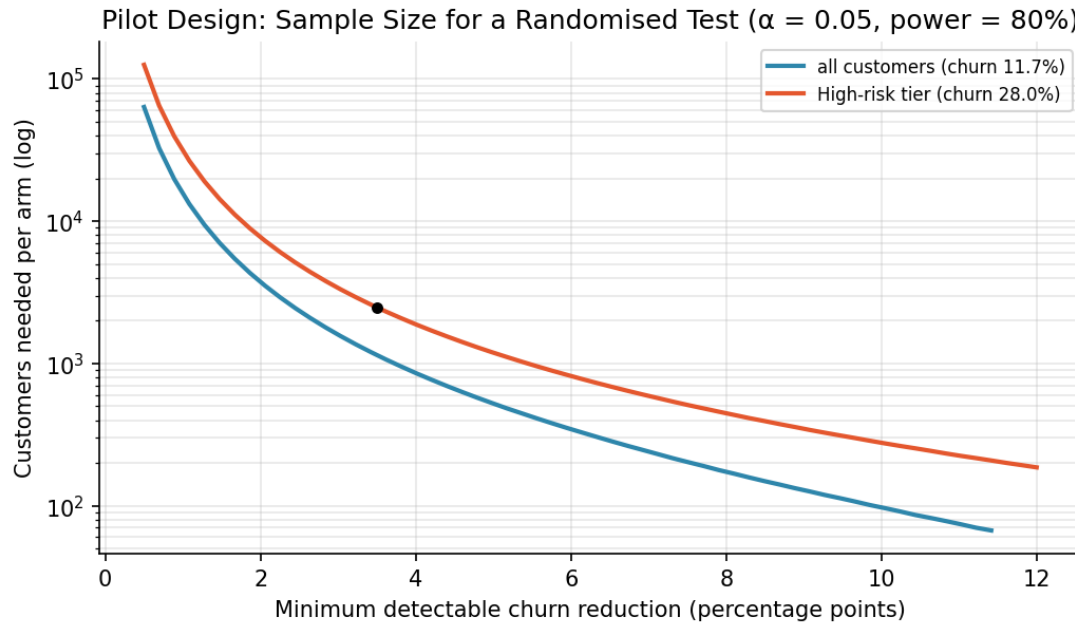


Figure J.3. Pilot Design: Sample Size for a Randomised Test ($\alpha = 0.05$, power = 80%)

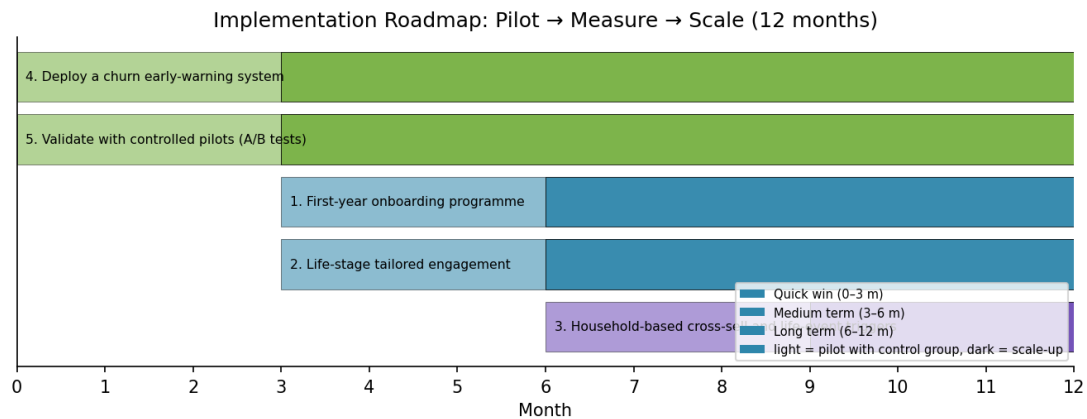


Figure J.4. Twelve-month implementation roadmap: each initiative is piloted against a control group before scale-up.

Table J.1. Model card (Mitchell et al., 2019).

Item	Description
Model	hill_climb (lgbm 0.50, lgbm_tuned 0.50)
Intended use	Rank active policyholders by 12-month churn risk to prioritise retention outreach and allocate retention budget.
Out-of-scope use	Not for premium pricing, underwriting, claims decisions or denial of service; not validated for other products/markets.
Training data	160,000 customers, 24 input features; target 'Churn' (churn = 1); churn rate 11.7%.
Excluded variables	acct suspd date year, acct suspd date month, acct suspd date (days since), acct suspd date (post-event leakage)
Validation	Stratified 5-fold CV (out-of-fold), target encoding inside folds, shuffled-label & adversarial checks, bootstrap & DeLong CIs.

Item	Description
Performance	ROC-AUC 0.702 (95% CI 0.698–0.707); PR-AUC 0.329; Brier 0.087; ECE 0.002; EMPC 4.31 USD/customer.
Calibration	isotonic (selected by cross-validated log-loss).
Decision rule	Risk tiers at the 50%/80% quantiles; campaign size chosen by expected profit (see Section 4).
Explainability	Global: SHAP, permutation importance, PDP, EBM shape functions (if used). Local: SHAP waterfall per customer for agents.
Fairness	Audited on age band, has children, marital status; max AUC gap 0.162.
Monitoring	Monthly: PSI per feature (alert > 0.25), score distribution, realised churn by tier, calibration; retrain quarterly or if AUC drops > 0.03.
Limitations	Observational snapshot data; causal effects assume no unmeasured confounding; ROI depends on stated cost/acceptance assumptions.

Table J.2. Fairness summary by attribute.

Attribute	Min sel	Max sel	Min auc	Max auc	Min cal	Max cal	Selection ratio	Auc gap
age band	0.173	0.543	0.596	0.758	0.971	1.225	0.318	0.162
has children	0.205	0.233	0.683	0.717	0.996	1.005	0.881	0.034
marital status	0.217	0.223	0.700	0.705	0.995	1.008	0.973	0.005

Table J.3. Pilot design: sample size per arm.

Intervention	Effect source	Baseline churn target segment	Expected reduction pp	N per arm	Total pilot size	Share of high tier %
Generic retention offer	assumed success rate × 50% (conservative)	0.280	3.499	2,481	4,962	75.4

Table J.4. Implementation roadmap.

Priority	Initiative	Phase	Start month	Pilot months	Scale months	Kpi
4	Deploy a churn early-warning system	Quick win (0–3 m)	0	3	3	Precision/recall of alerts; retained revenue vs control group
5	Validate with controlled pilots (A/B tests)	Quick win (0–3 m)	0	3	3	Uplift in retention rate (treated – control)
1	First-year onboarding programme	Medium term (3–6 m)	3	3	3	13th-month persistency ratio; first-year churn rate
2	Life-stage tailored engagement	Medium term (3–6 m)	3	3	3	Churn rate by age band; adoption of tailored offers
3	Household-based cross-sell and life-event triggers	Long term (6–12 m)	6	3	6	Products per household; churn after life events

Table J.5. Full recommendation list with evidence and KPIs.

Priority	Recommendation	Action	Evidence	Estimated impact	Kpi
1	First-year onboarding programme	Welcome call, policy walkthrough, app activation and 90-day check-in for new customers; first renewal treated as a critical moment.	'0-6m' churns at 48.8% (4.2x avg); IV 0.89; OR 12.20; SHAP rank #10		13th-month persistency ratio; first-year churn rate
2	Life-stage tailored engagement	Tailor products, pricing communication and service channels to the highest-risk age groups (e.g. digital-first onboarding and flexible payment for younger customers; loyalty recognition for long-standing older customers).	'25-34' churns at 14.6% (1.2x avg); IV 0.04; OR 0.92; HR 0.48; SHAP rank #16		Churn rate by age band; adoption of tailored offers
3	Household-based cross-sell and life-event triggers	Use household profile and life events (moving, marriage, children) as triggers for coverage reviews and multi-policy bundles.	'0-1' churns at 13.5% (1.2x avg); IV 0.01; SHAP rank #4		Products per household; churn after life events
4	Deploy a churn early-warning system	Score all active policies monthly with this model; route High-risk/Priority-Save customers to retention teams, lower tiers to automated digital nudges; refresh the model quarterly.	Model ROC-AUC 0.702; top-20% captures 51% of churners	optimal campaign: target top 10% → net 590.81K	Precision/recall of alerts; retained revenue vs control group
5	Validate with controlled pilots (A/B tests)	For each intervention keep a random control group (e.g. 20% of targeted customers) to measure true uplift before scaling.	What-if results are associational		Uplift in retention rate (treated – control)

Appendix K. Reproducibility

All results were produced by one notebook executed top-to-bottom (Restart & Run All) with random seed 42, 5-fold stratified cross-validation and the settings in Appendix B. Software: Python 3.11.9, pandas 2.3.3, scikit-learn 1.6.1, LightGBM 4.7.0, XGBoost 3.2.0, CatBoost 1.2.10, SHAP 0.51.0, statsmodels 0.15.0, lifelines 0.30.3, Optuna 4.9.0; platform Windows 10. The submission package contains the executed notebook with all outputs, every figure and table (CSV and Excel), the model card, the per-customer action list and an interactive dashboard.

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